

On-Chip LC filters for Hybrid Spin-Superconducting Systems

Oswin Cervantes

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**Department of Electrical Engineering and Computer Science,
MIT Research Laboratory of Electronics**

Group Meeting

28 July 2026

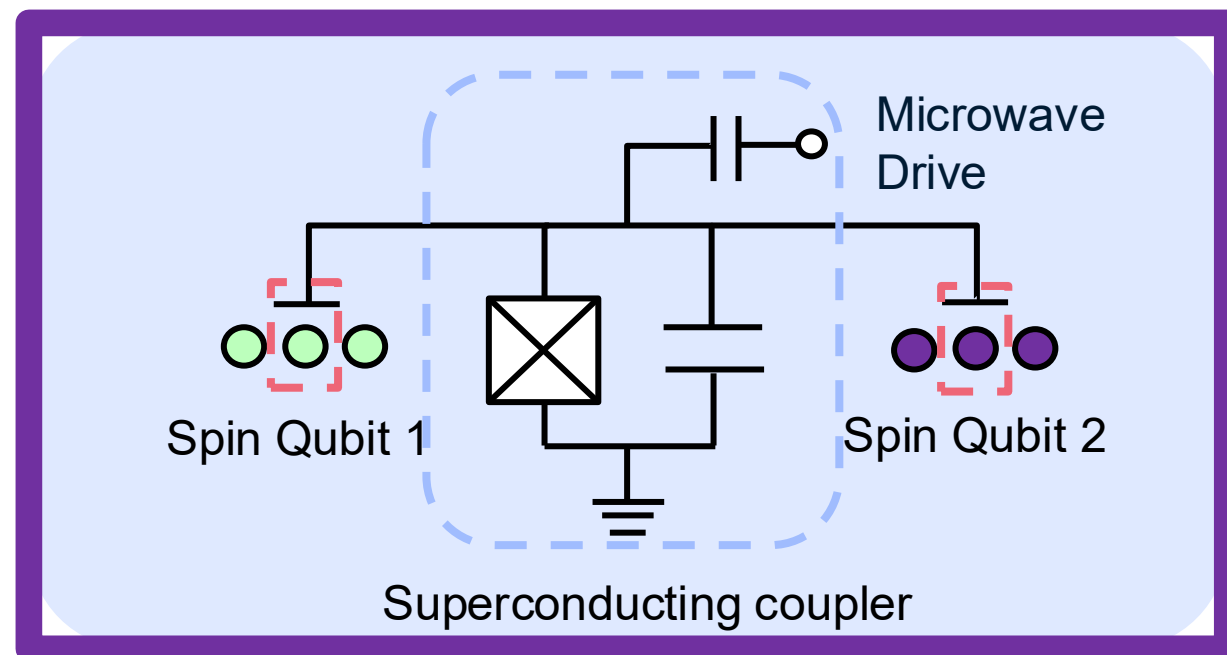
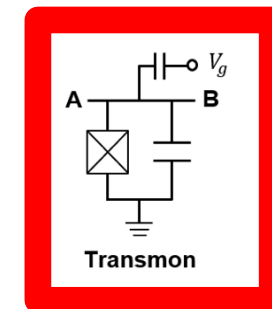
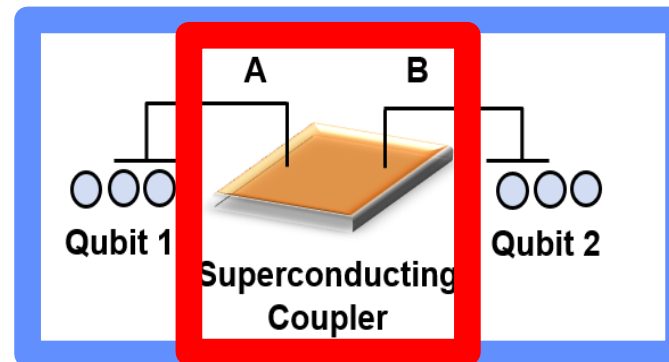
Why do we want to couple spin and superconducting qubits?

Spin qubits:

- Conventional CMOS
- Compact footprint
- T1 up to seconds.

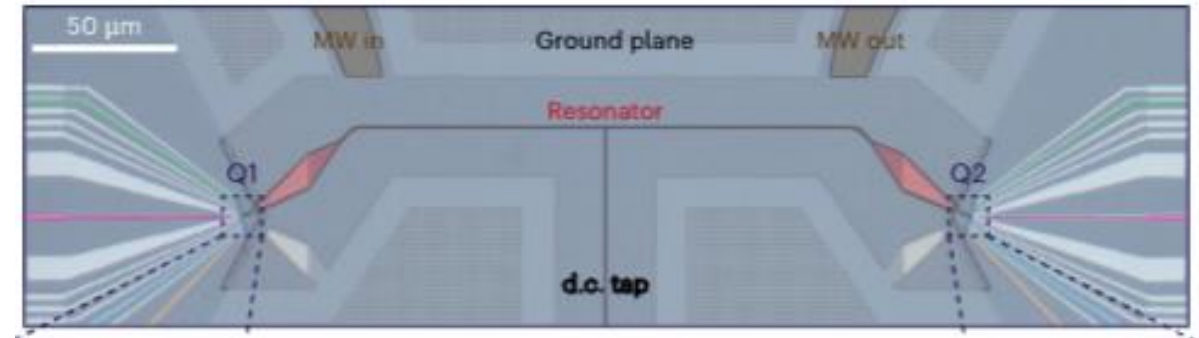
Superconducting add fast microwave control and longer-range connectivity. Allows room for electronic gate lines

Note: Superconducting qubits in our case are not information storing, but solely coupling devices



Current methods: spin-spin coupling to resonator both have their own advantages and challenges.

- 250- μm NbTiN (Niobium-Titanium) half-wave resonator
- 6.91 GHz
- 3 k Ω Impedance
- Temperature of 8 mK
- $Q \sim 3800$
- Mediates virtual-photon spin coupling and enables dispersive readout.

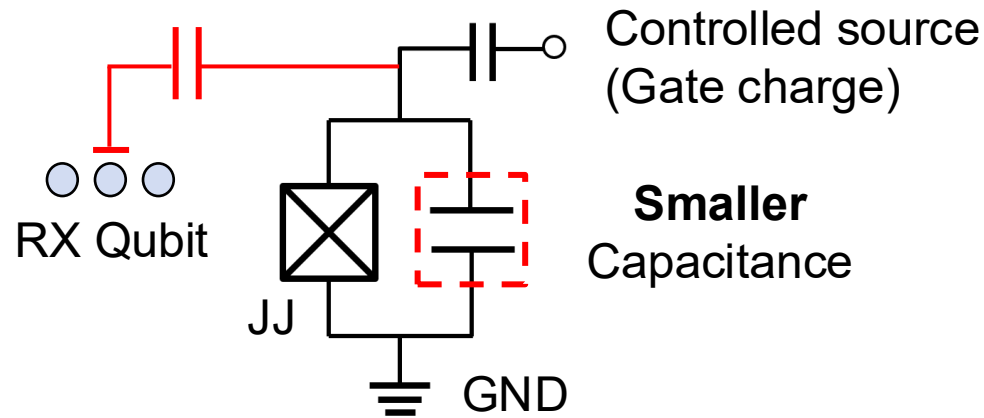


Vandersypen et al.

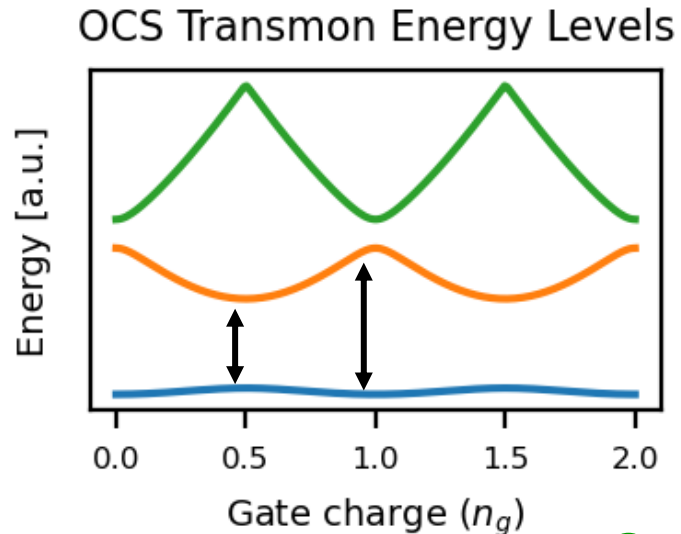
Main Take Away:

1. High coupling - high impedance leads to high photon loss through gate line

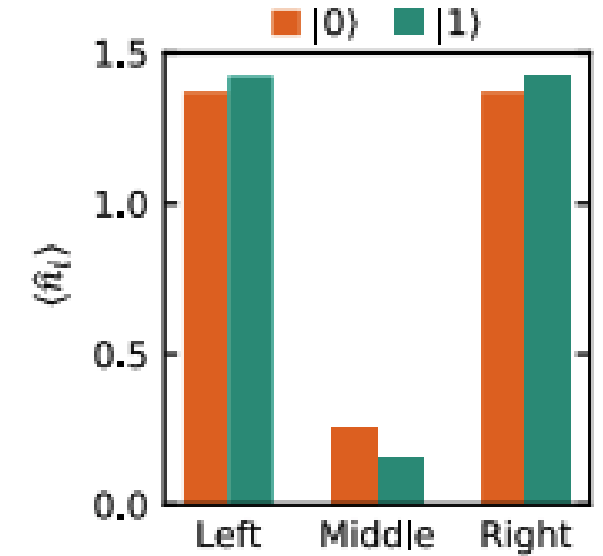
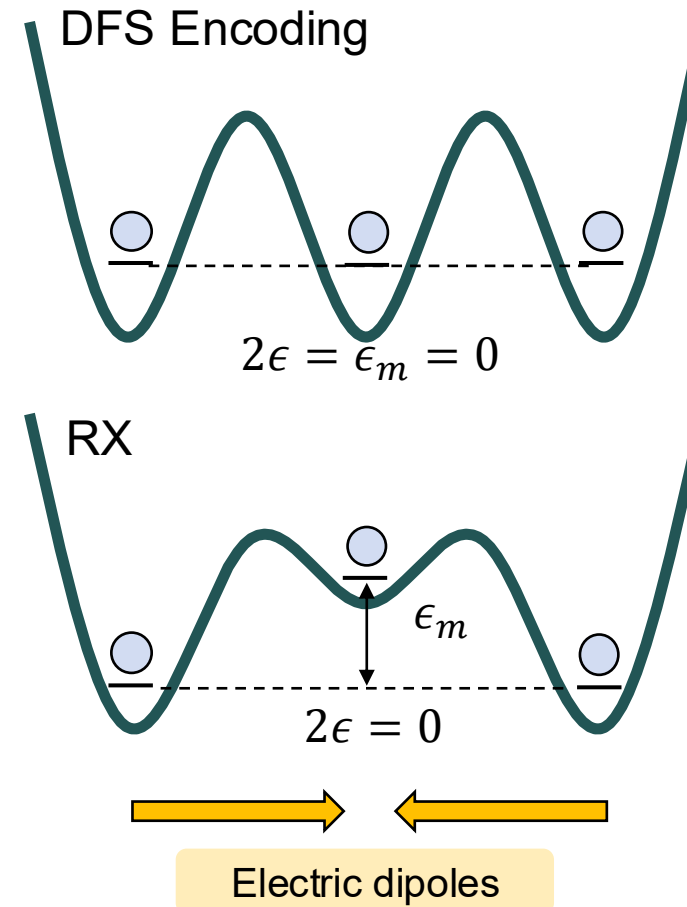
OCS Transmon: offset-charge-sensitive transmon



Dependent on gate charge



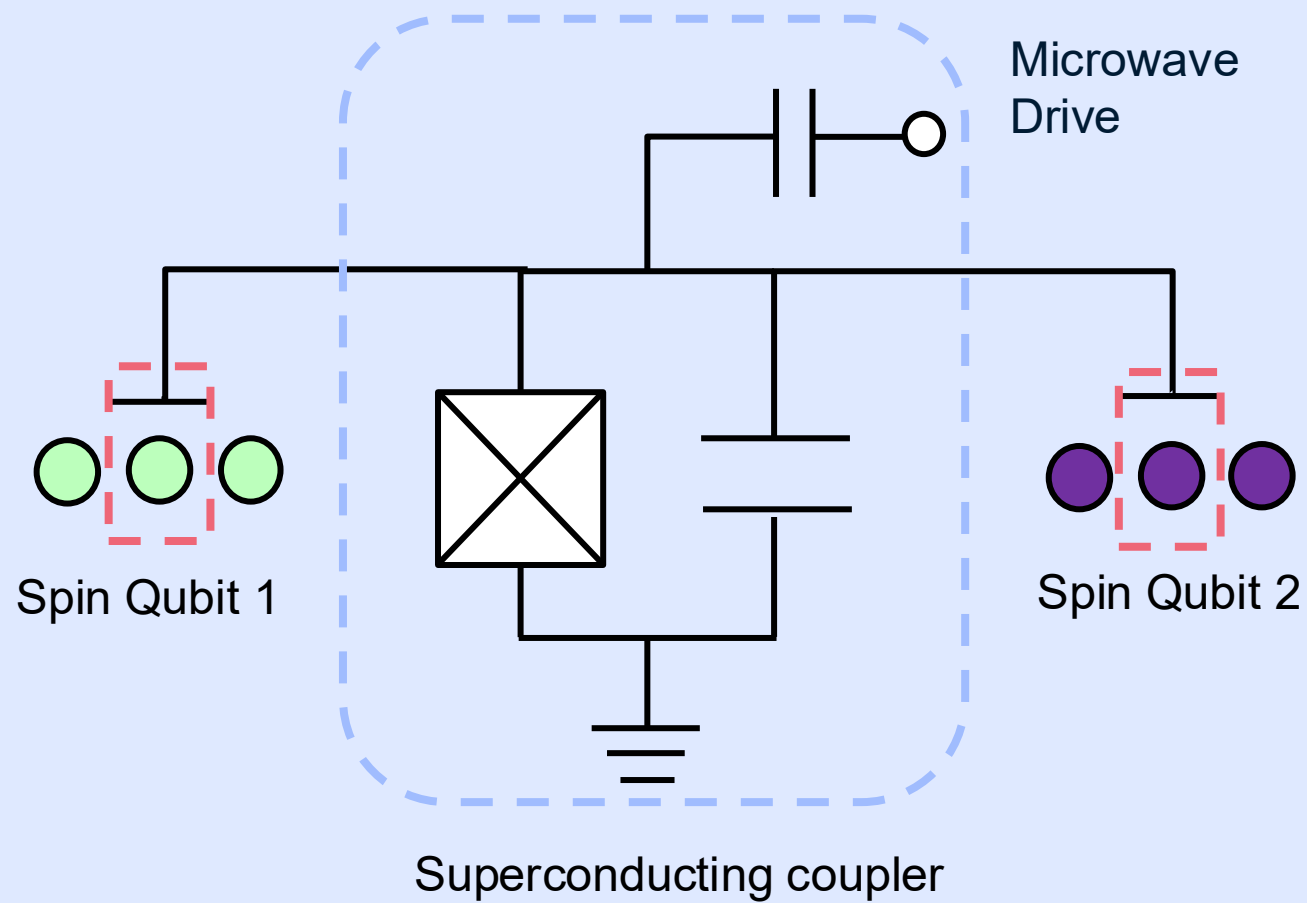
EO Qubits

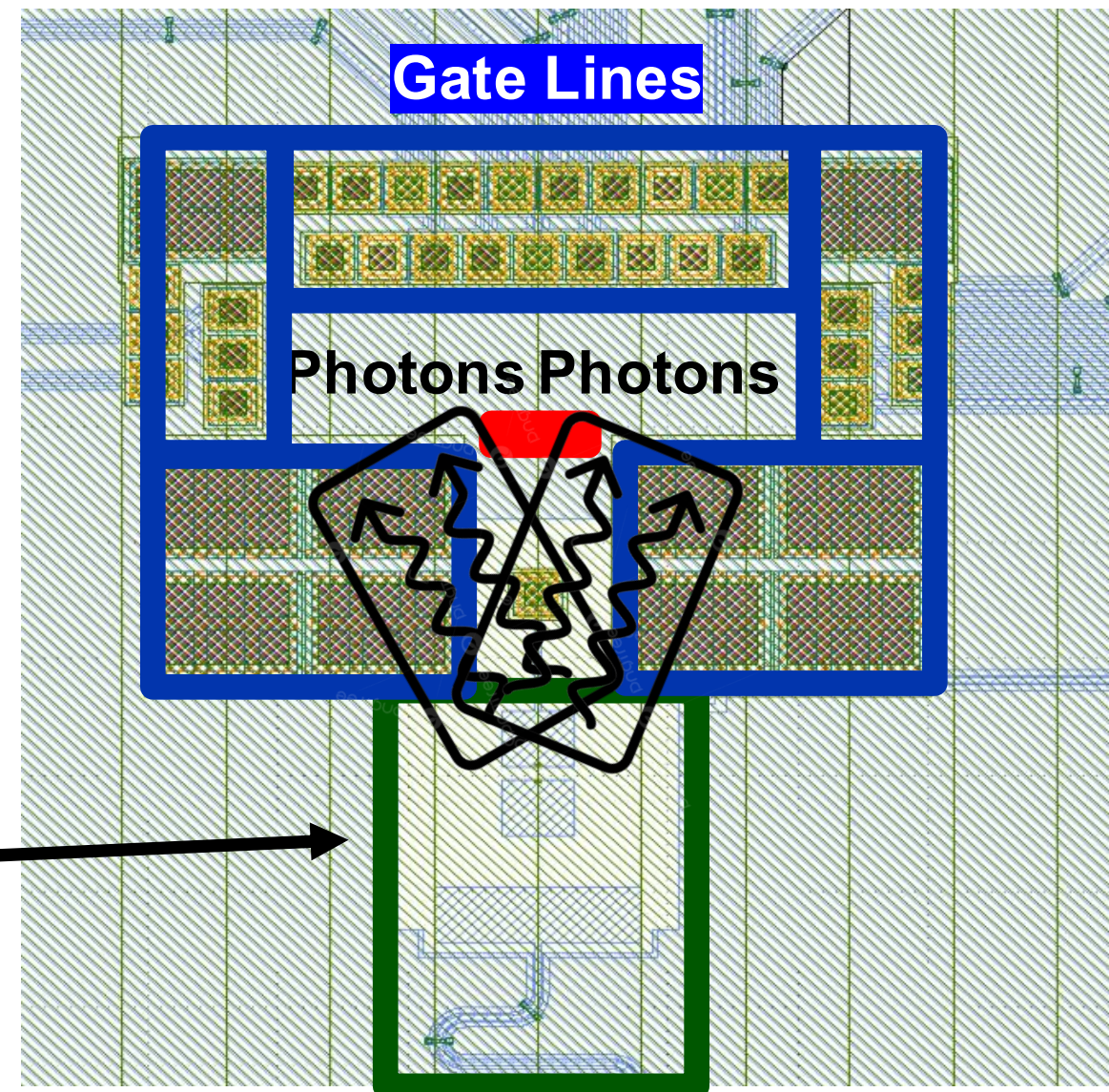
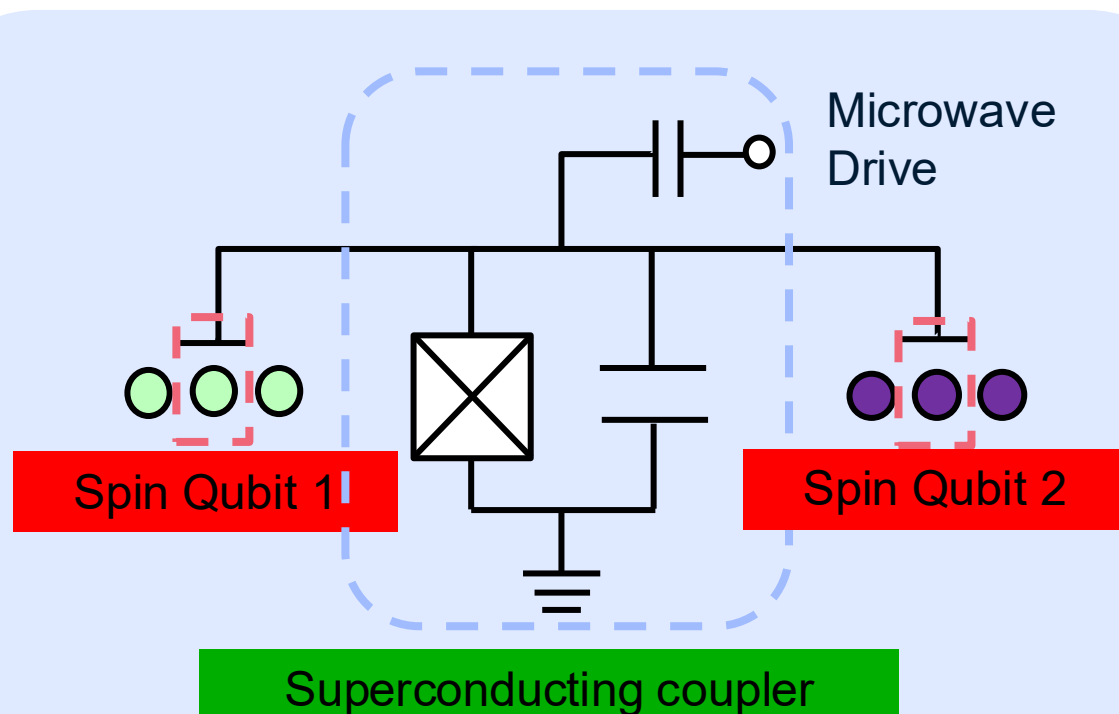


Kang et al.

Read Harry Kang's Paper!

Qubit states capacitively change the resonance frequency





3-5 GHz
Operation

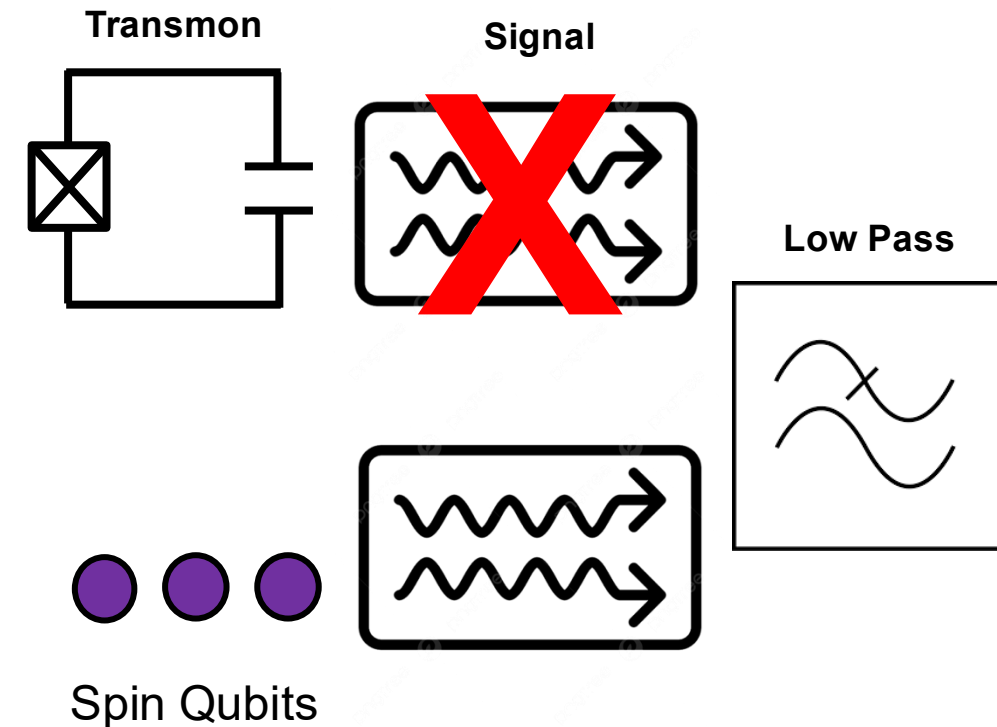
Design **filters** that blocks transmission of during the operation range.

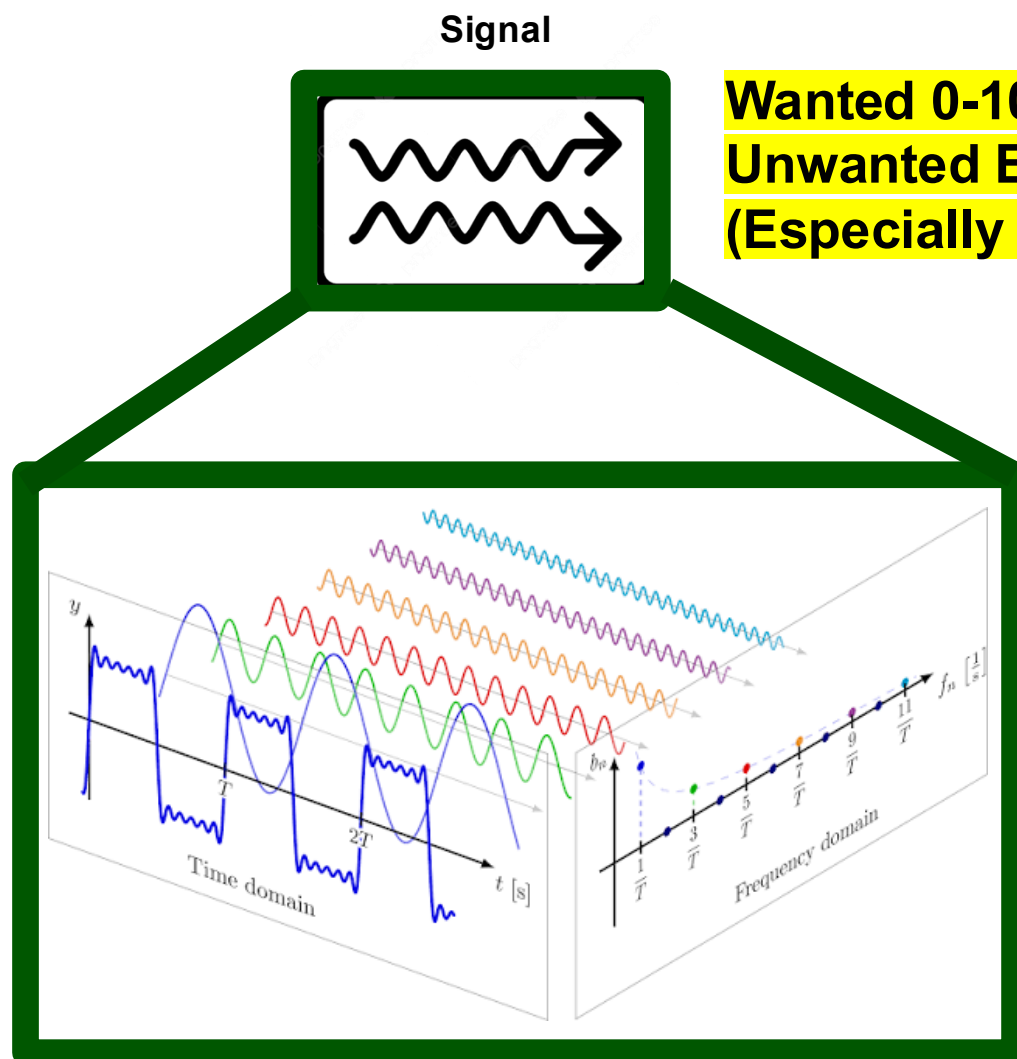
Operation Range:

- OCS Transmon: 3~5 GHz - **STOP**
- Quantum Dots: 0-100 MHz - **ALLOW**

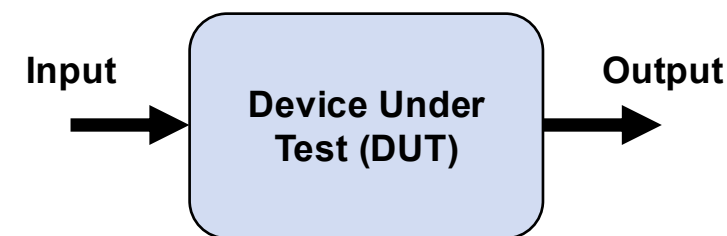
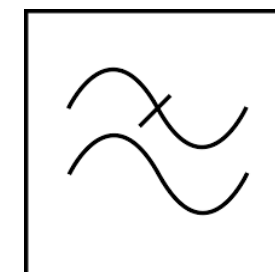
Constraints:

- Manufacturability (Fabrication)
- DC Filter

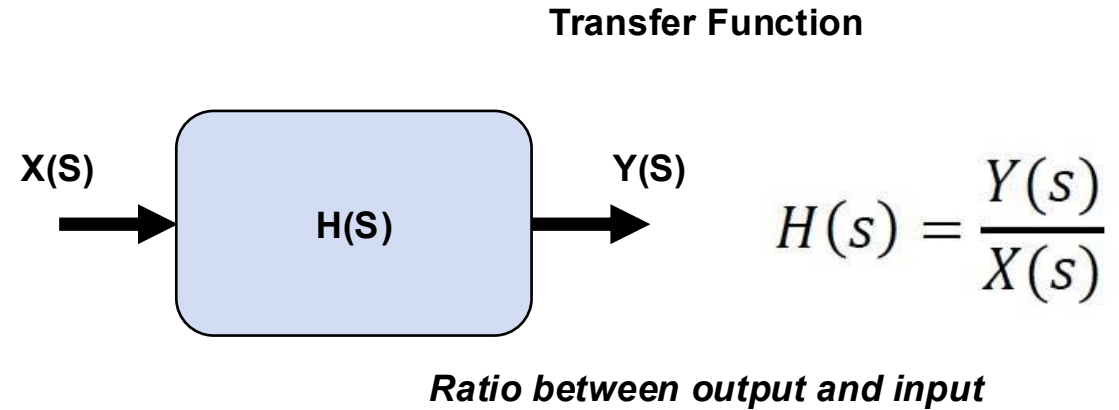
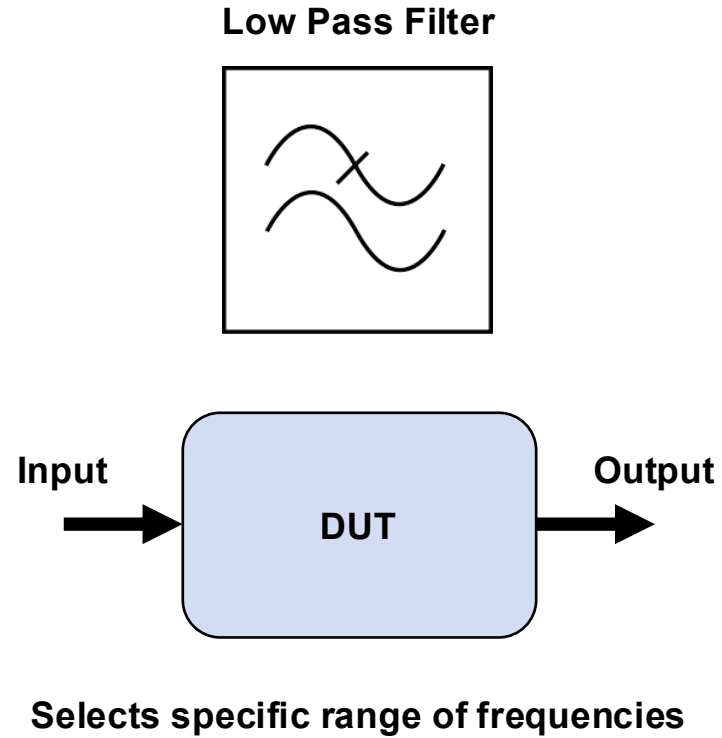




Low Pass

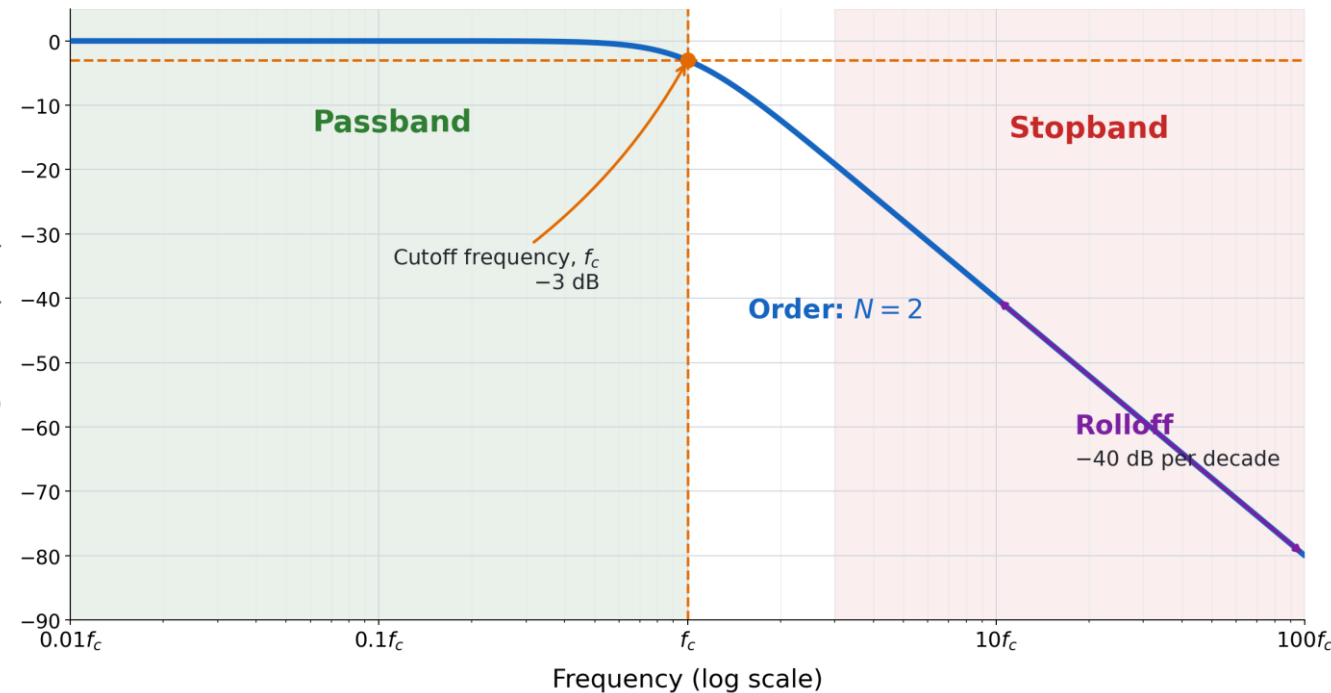


Selects specific range of frequencies



Transfer Function (S_{11}/S_{21}) --- Frequency Response

Anatomy of a Low-Pass Filter Response
Second-order Butterworth example



Sometimes in log scale for large ranges

Important terms:

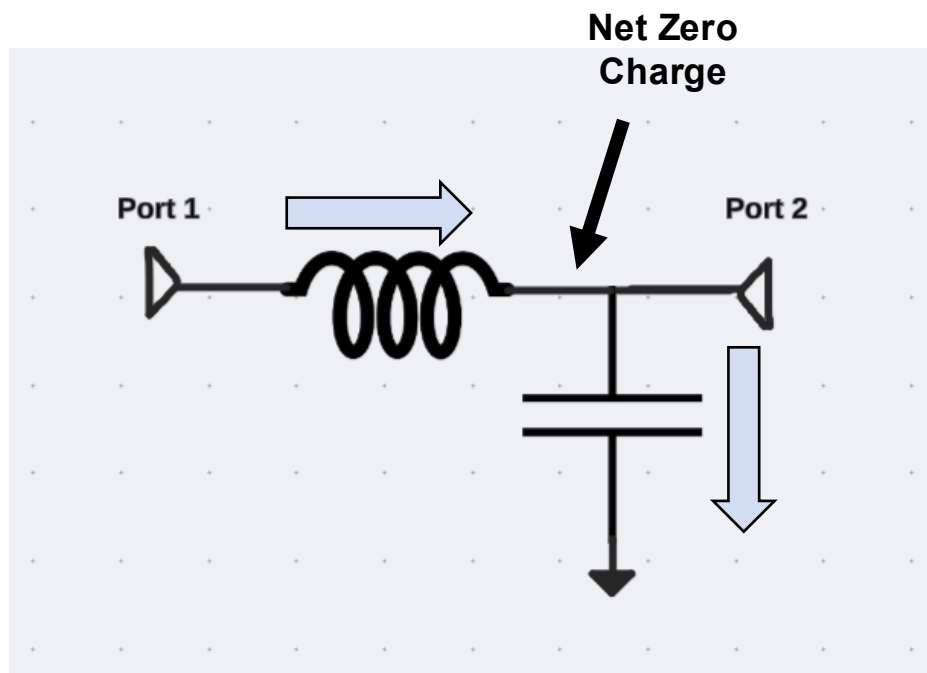
Cutoff frequency – The frequency where -3dB or half power is reached

Rolloff – The attenuation factor per decade

Order – The leading exponential factor in the **denominator**, which leads to $-20 * N$ dB per decade

Passband - Frequency region where signal minimally attenuated

Stopband – Frequency region where signal is attenuated



Using basic conservation of charge to find relationship between V_{in} and V_{out}

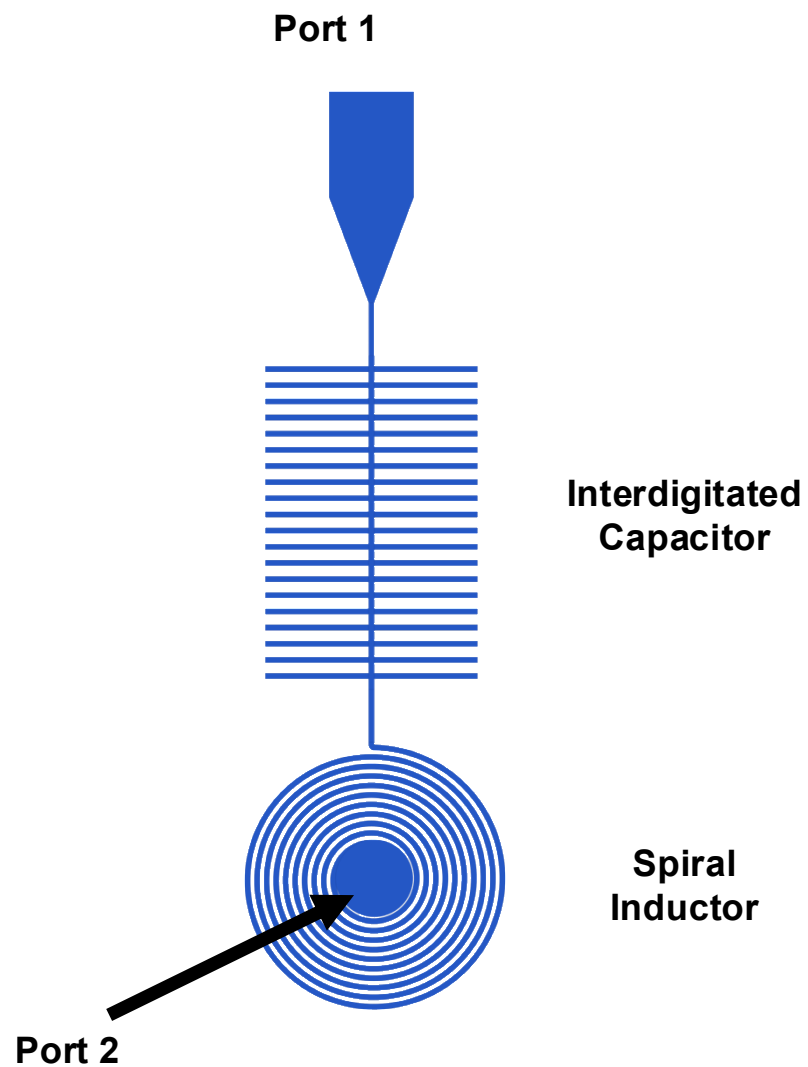
$$1. \frac{V_{out}}{\frac{1}{j\omega C}} + \frac{V_{in} - V_{out}}{j\omega L} = 0 \text{ (no charge accumulation)}$$

$$2. V_{out} = \frac{1}{1 - \omega^2 LC} V_{in}$$

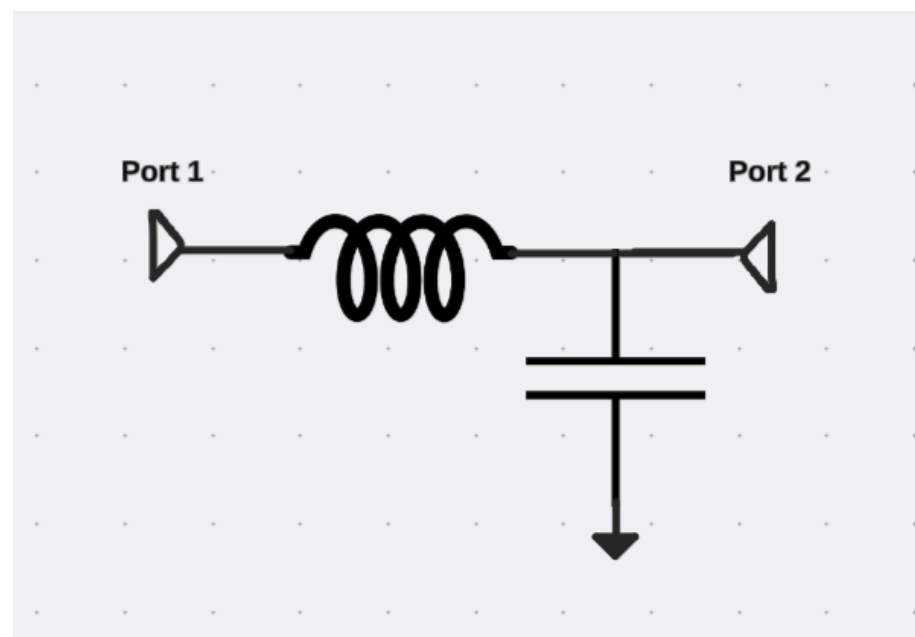
$$3. \frac{V_{out}}{V_{in}} = \frac{1}{1 - \omega^2 LC} \text{ (Frequency Response!)}$$

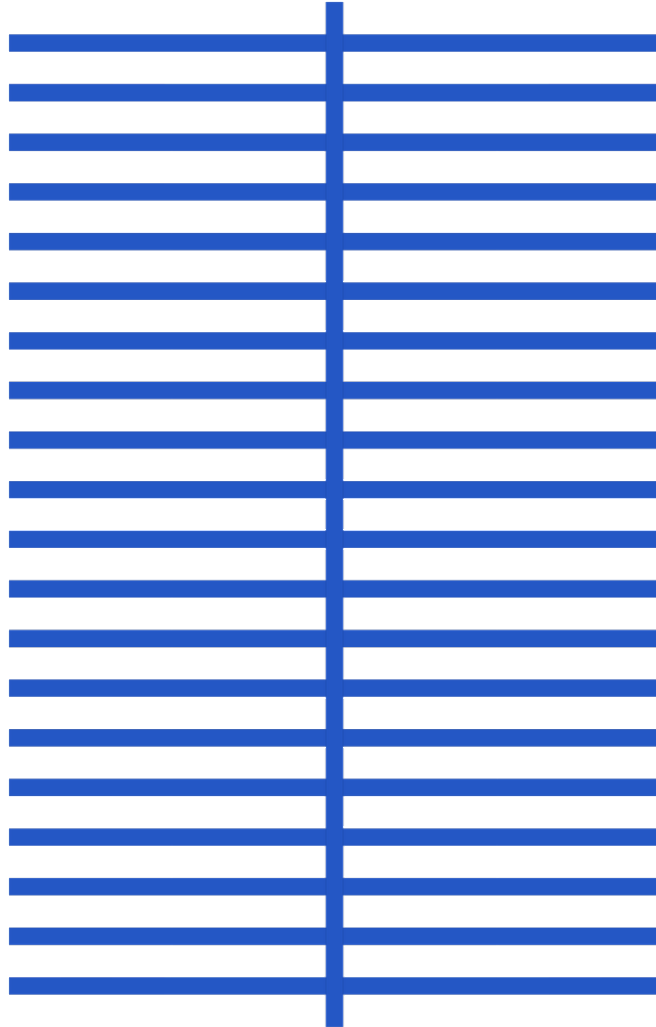
SECOND ORDER LOW PASS (IDEAL)

$$\text{Resonance: } \omega = \frac{1}{\sqrt{LC}}$$



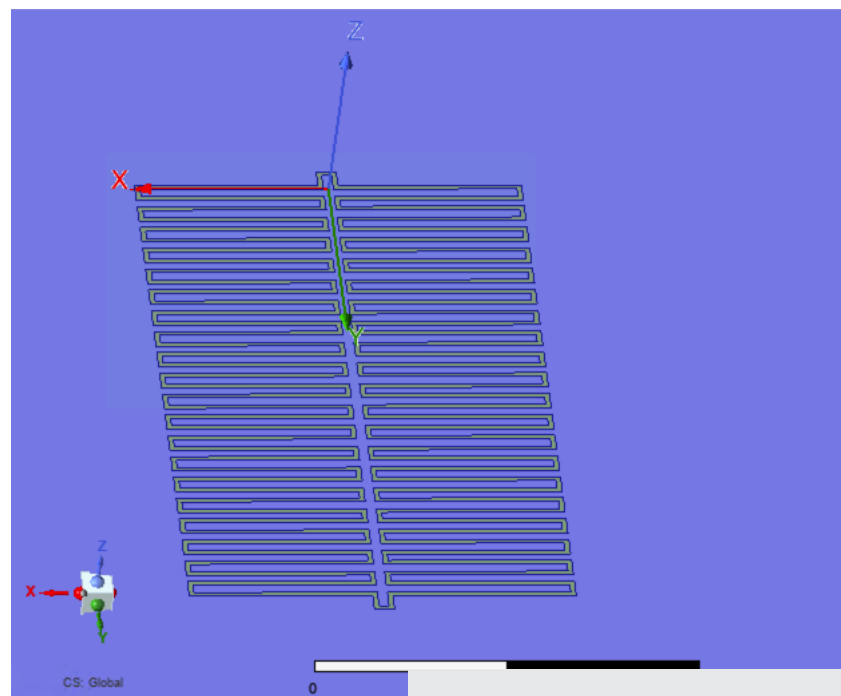
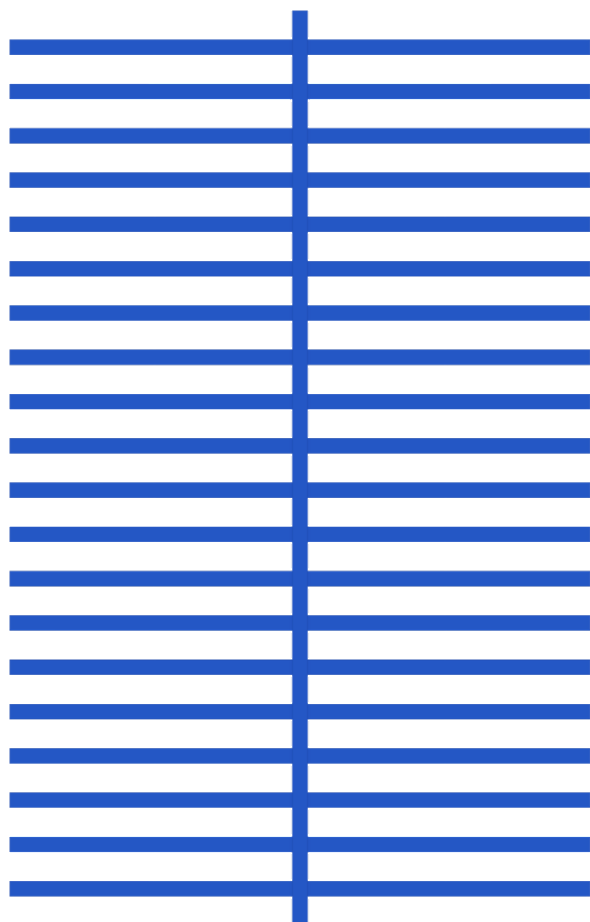
Lumped Element Second Order Low Pass





Interdigitated Capacitor – Parameters (IDC)

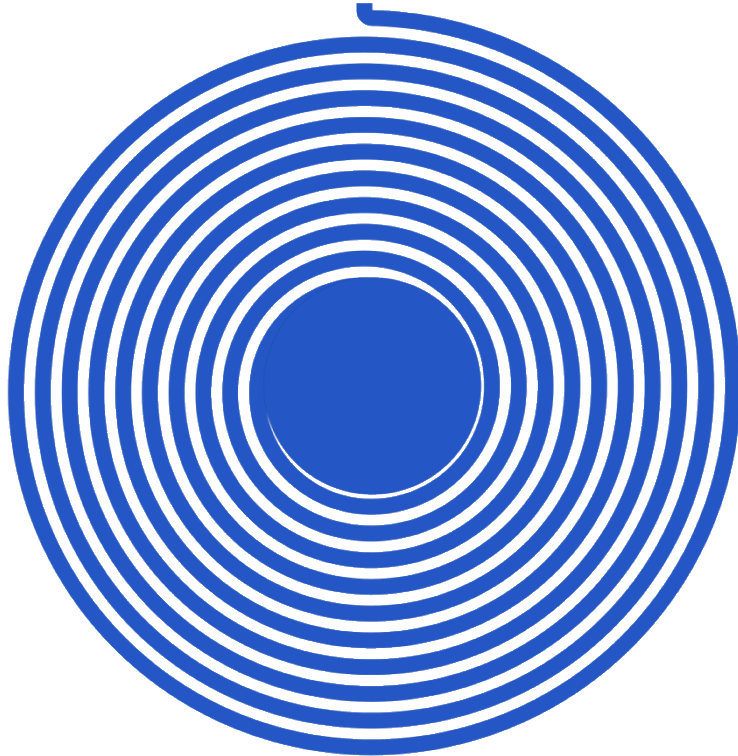
- Finger width
- Finger length
- Number of fingers
- Finger-to-finger spacing
- Ground-plane clearance
- Entrance width
- Fabrication bias



**Benchmark through Ansys
Maxwell, or approximate
through unit scaling of a
single capacitance block**



Spiral Inductor - Parameters



- Trace width

- Number of turns

- Turn pitch

- Trace-to-trace spacing

- Ground-plane clearance

- Inner radius

- Connection extension length

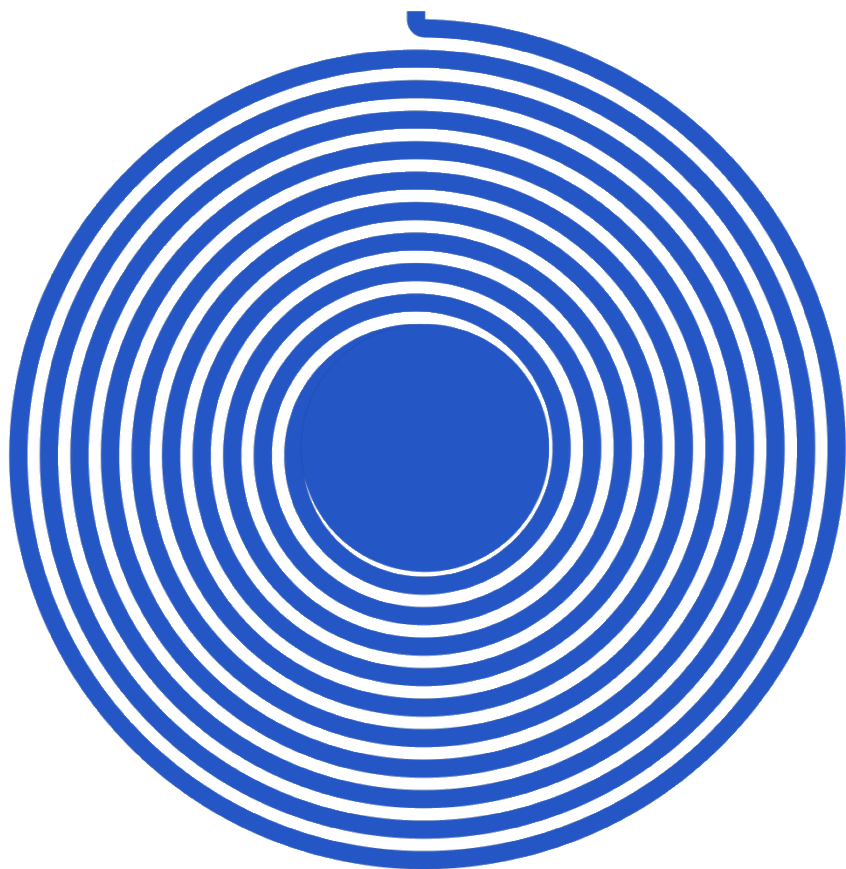
- Extension angle

- Bend radius

- Taper dimensions

- Geometry tolerance

- Fabrication bias



Circular planar spiral. The Mohan current-sheet approximation gives the geometric inductance as

$$L = \frac{\mu_0 N^2 d_{\text{avg}}}{2} \left[\ln \left(\frac{2.46}{\rho} \right) + 0.20 \rho^2 \right] \quad (1)$$

with

$$d_{\text{avg}} = \frac{d_{\text{out}} + d_{\text{in}}}{2}, \quad \rho = \frac{d_{\text{out}} - d_{\text{in}}}{d_{\text{out}} + d_{\text{in}}}. \quad (2)$$

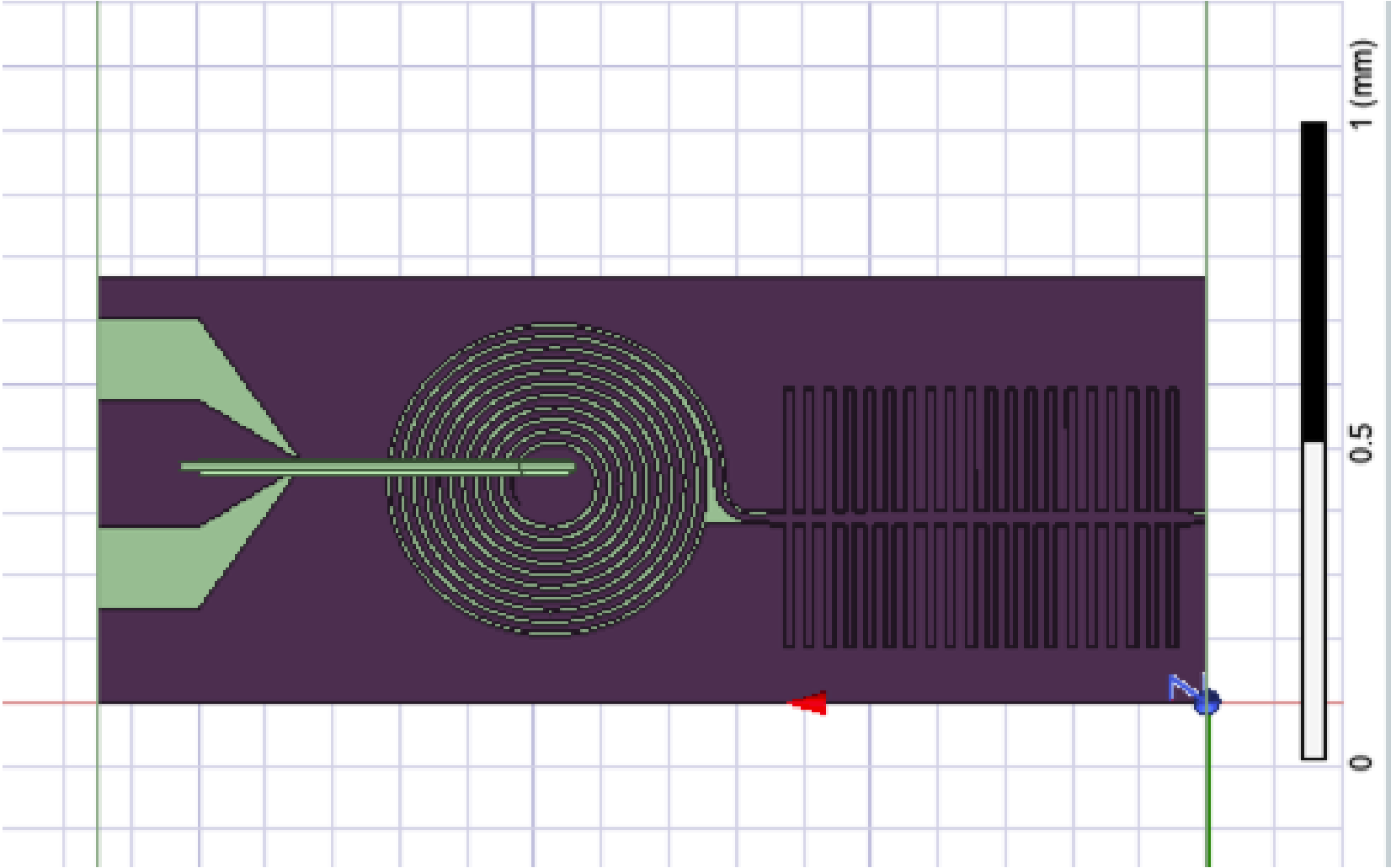
where:

L	geometric inductance, in henries;
N	number of spiral turns;
d_{out}	outer diameter;
d_{in}	inner diameter;
d_{avg}	average diameter;
ρ	dimensionless fill ratio;
μ_0	permeability of free space, $4\pi \times 10^{-7} \text{ H m}^{-1}$.

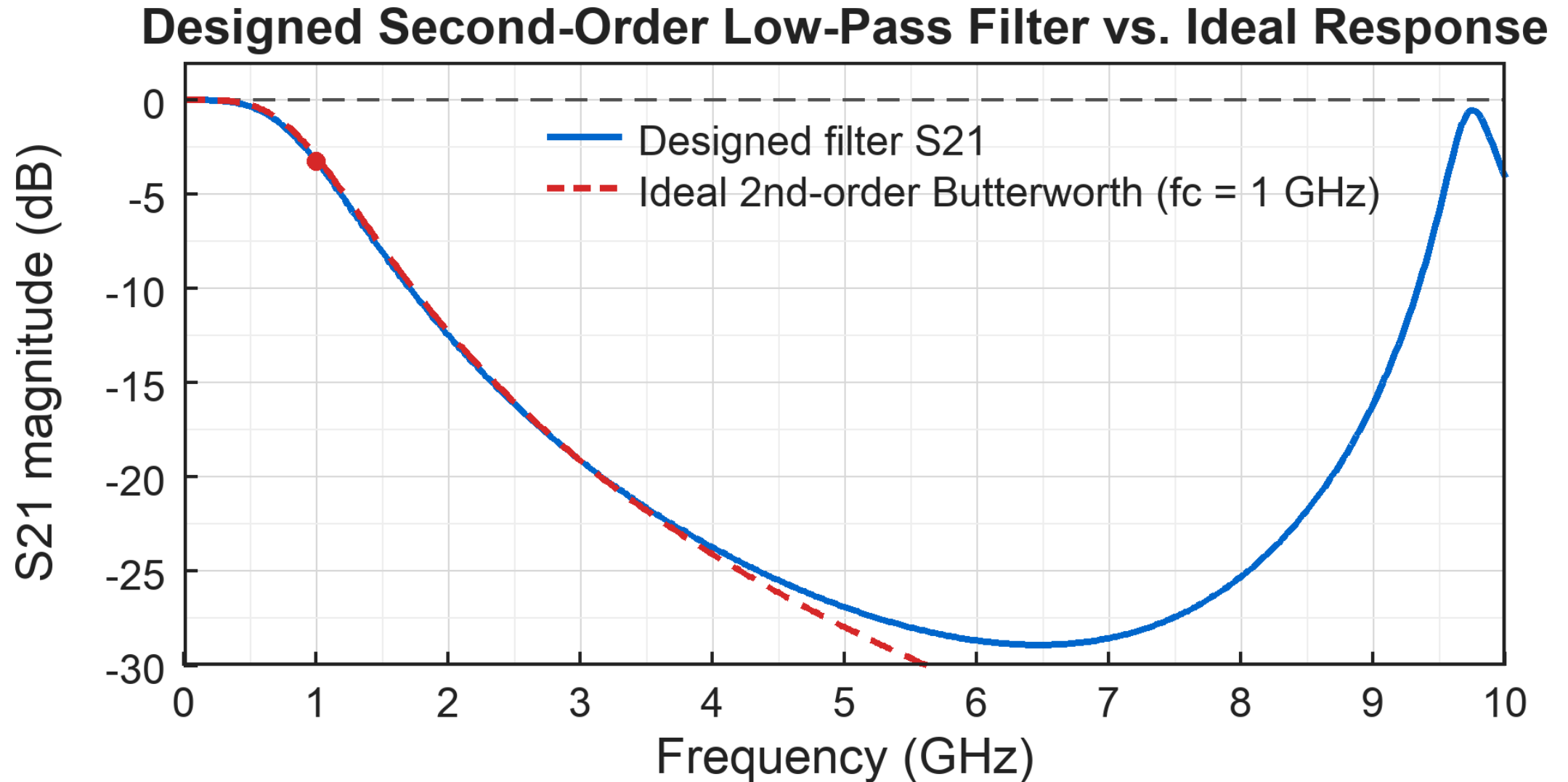
There is a great calculator created by Stanford for rough order of magnitude approx.

Inductance can also be approximated using the resonance frequency

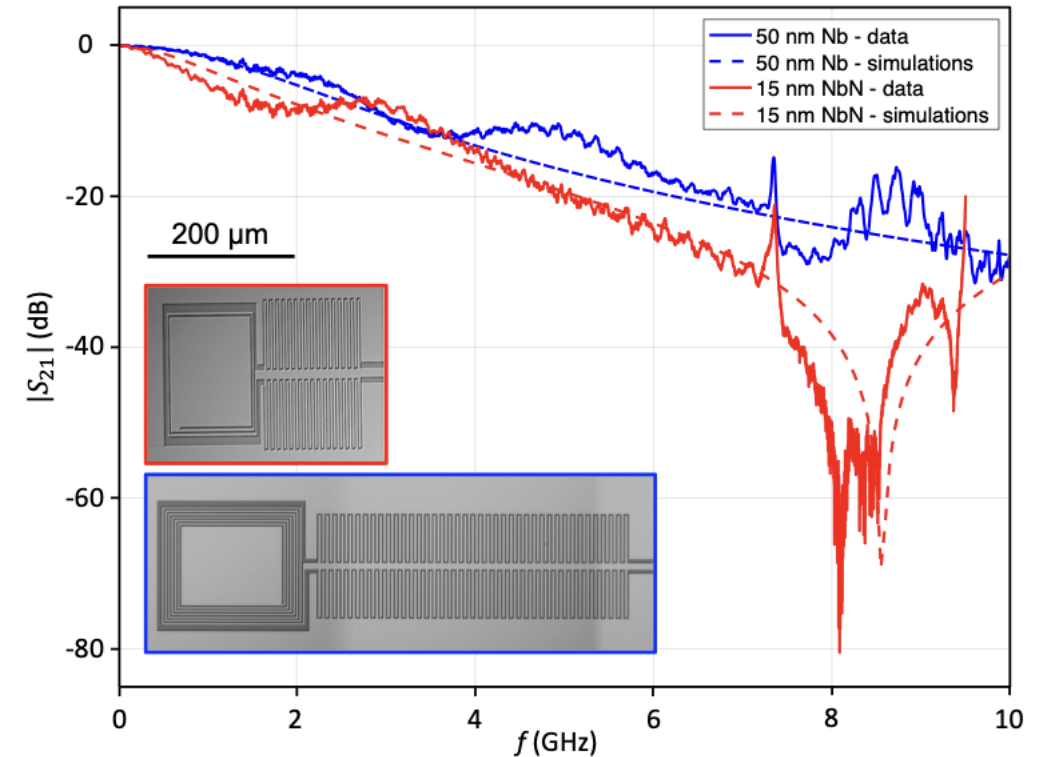
LC Filter Simulation - Setup



LC Filter Simulation

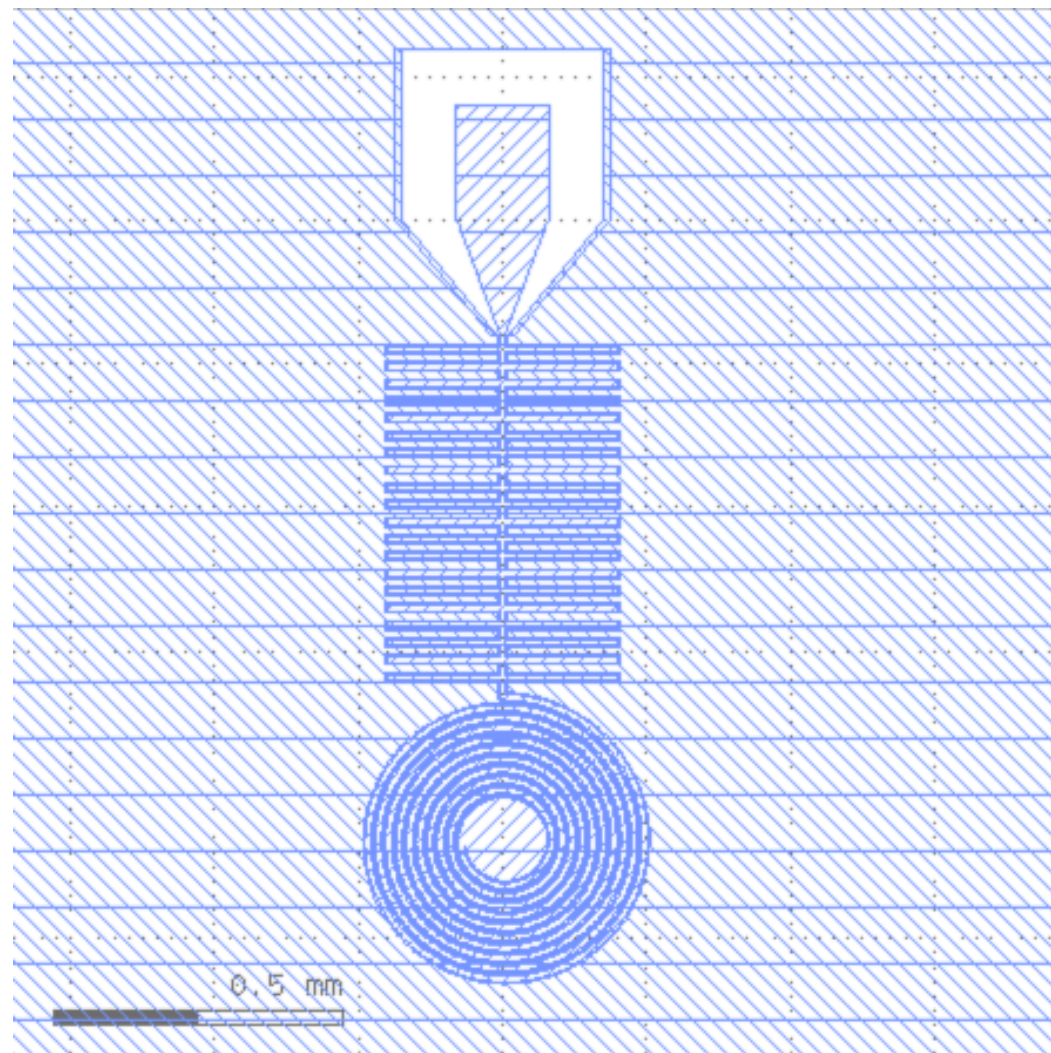


- Measured S_{21} : solid lines; simulated response: dashed lines
- **15-nm NbN filter:** ~60 dB attenuation near 8 GHz
- **50-nm Nb filter:** ~24 dB attenuation with $\sim 2\times$ greater length
- Insets show optical images of both fabricated filters

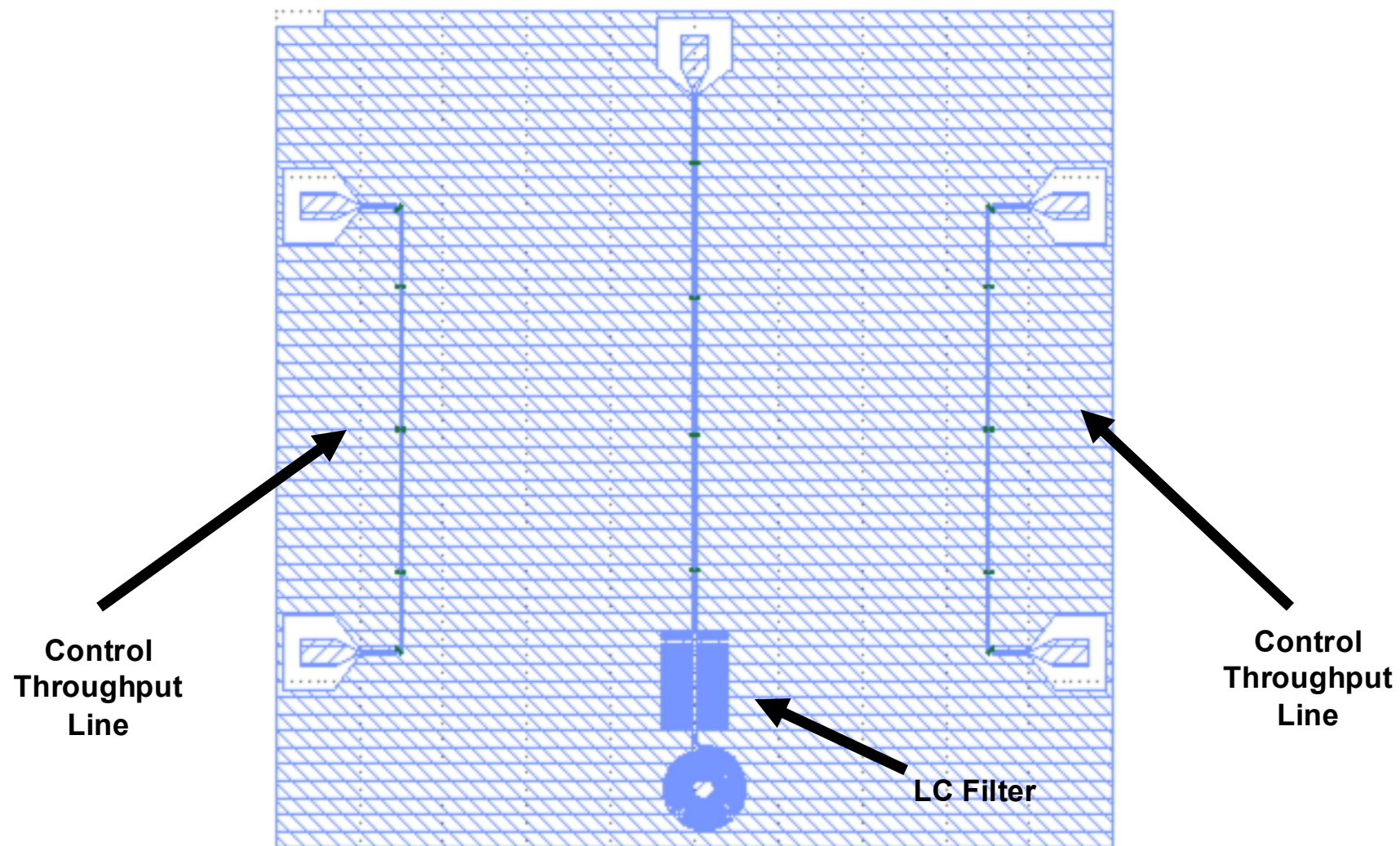


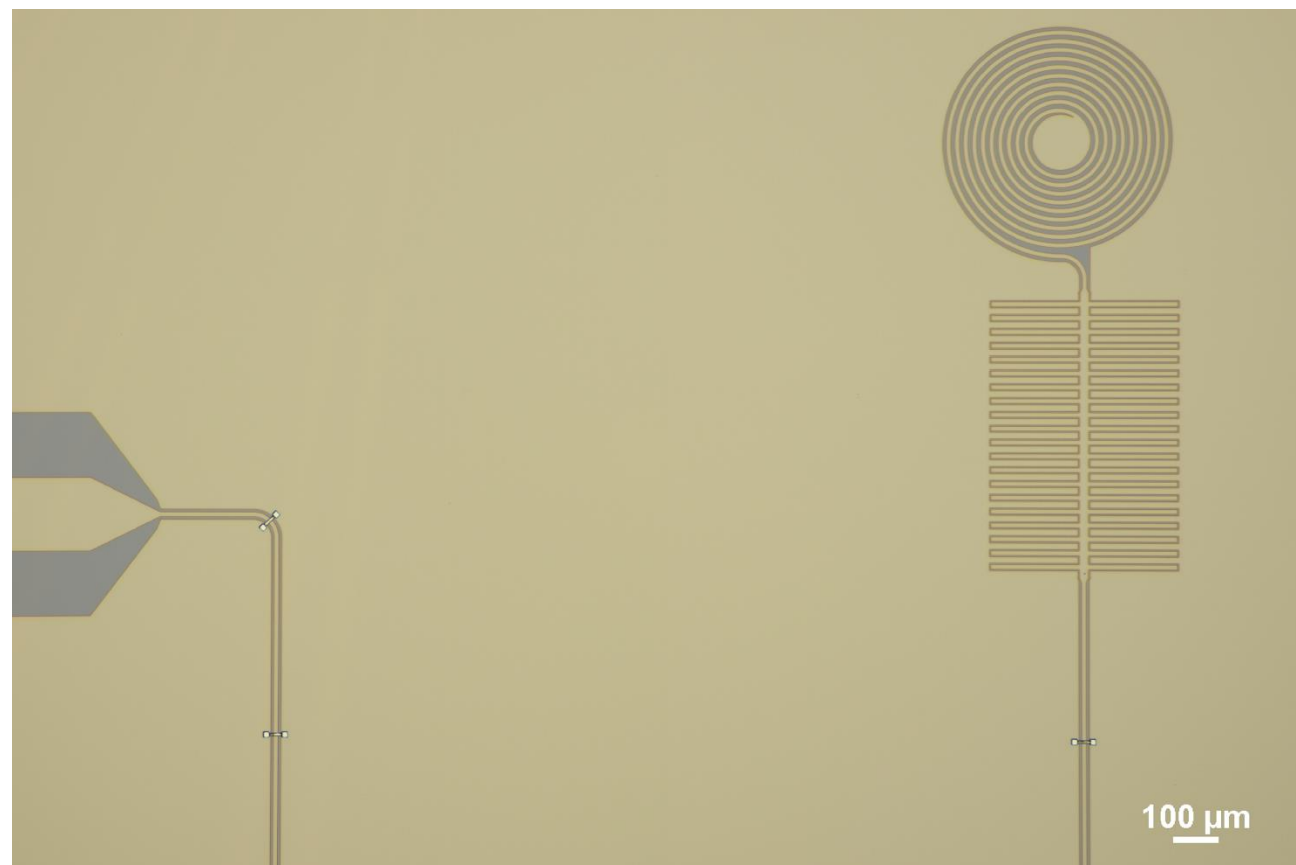
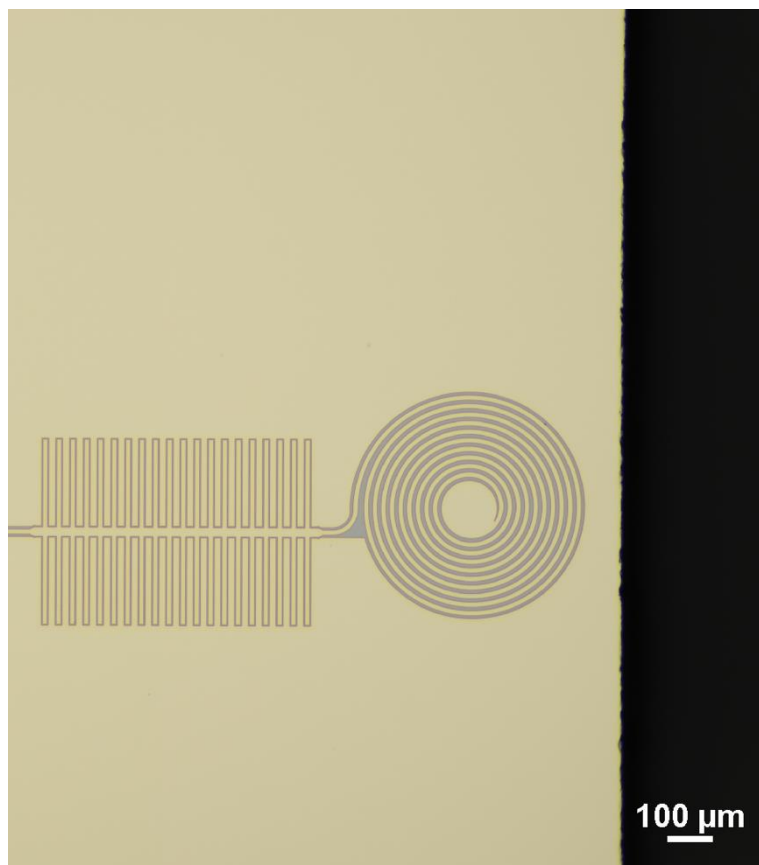
J.R. Petta et al.

Our Filters will be fabricated out of Nb (Low to Zero Kinetic Inductance)



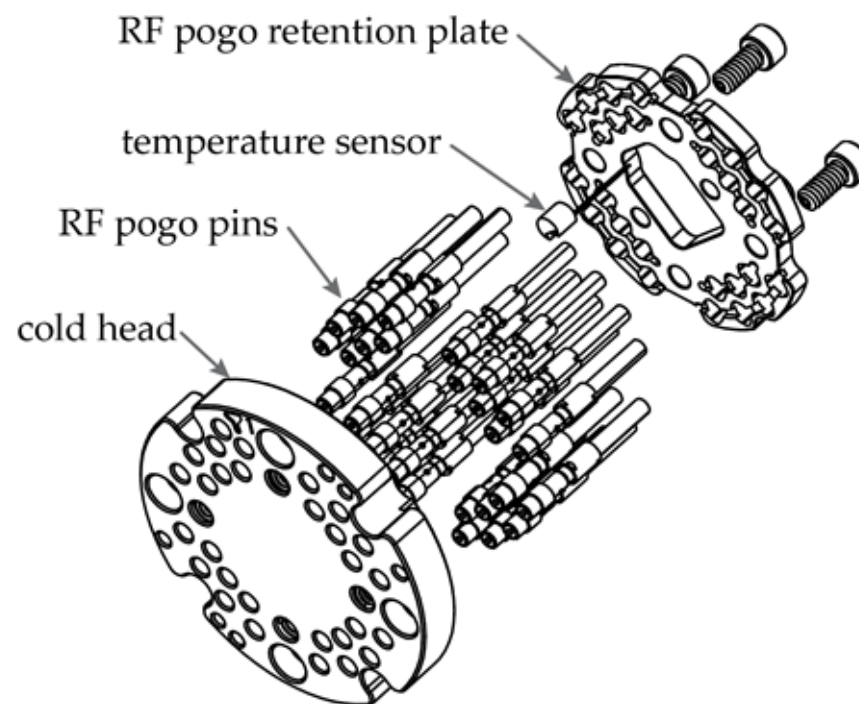
Experimental Design





Thank you, Fabrication Team!
Harry, Gabriel, Junyoung, Farid, and Chia-Chin

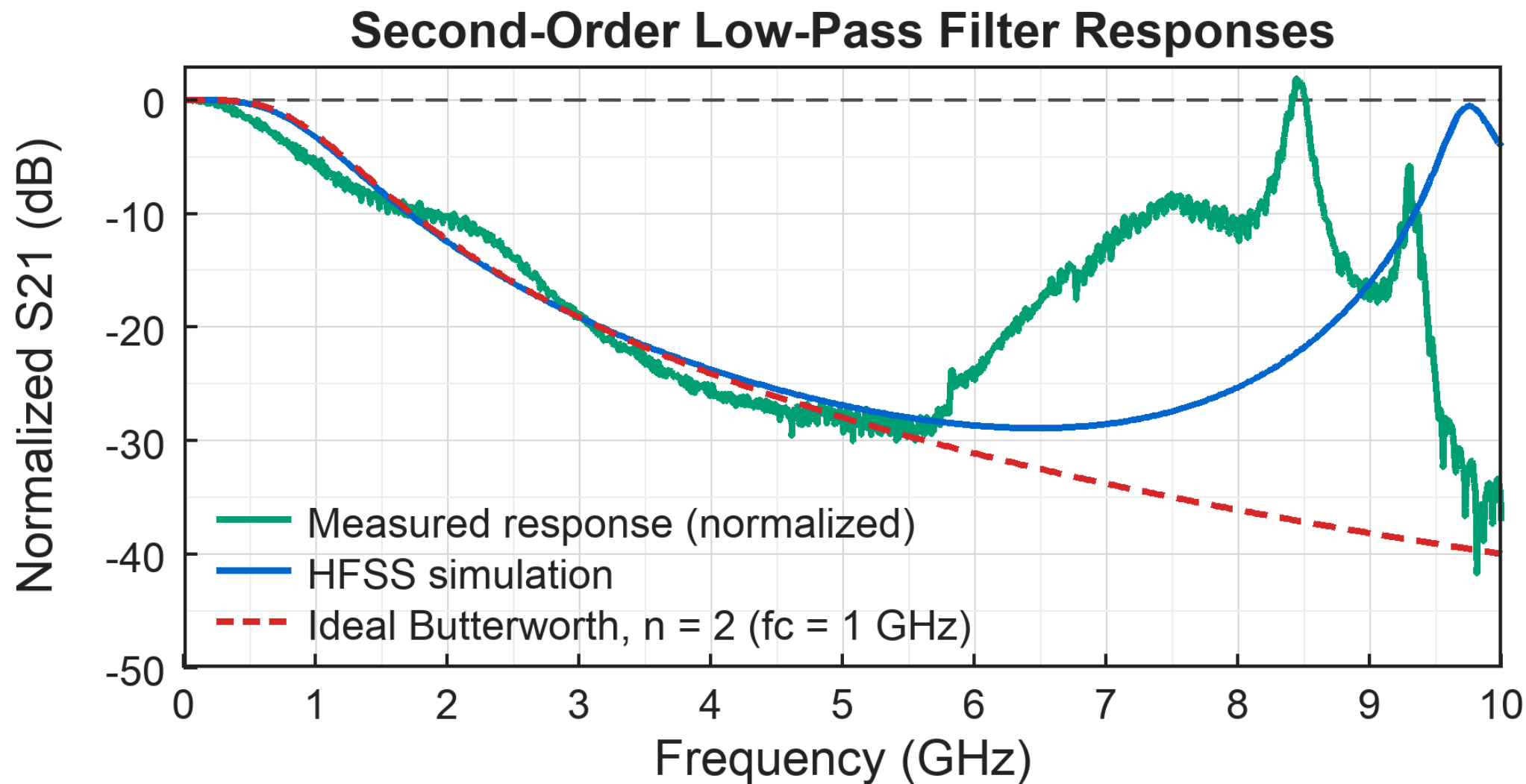
Thank you, Reed Foster
From Karl Berggren's Group



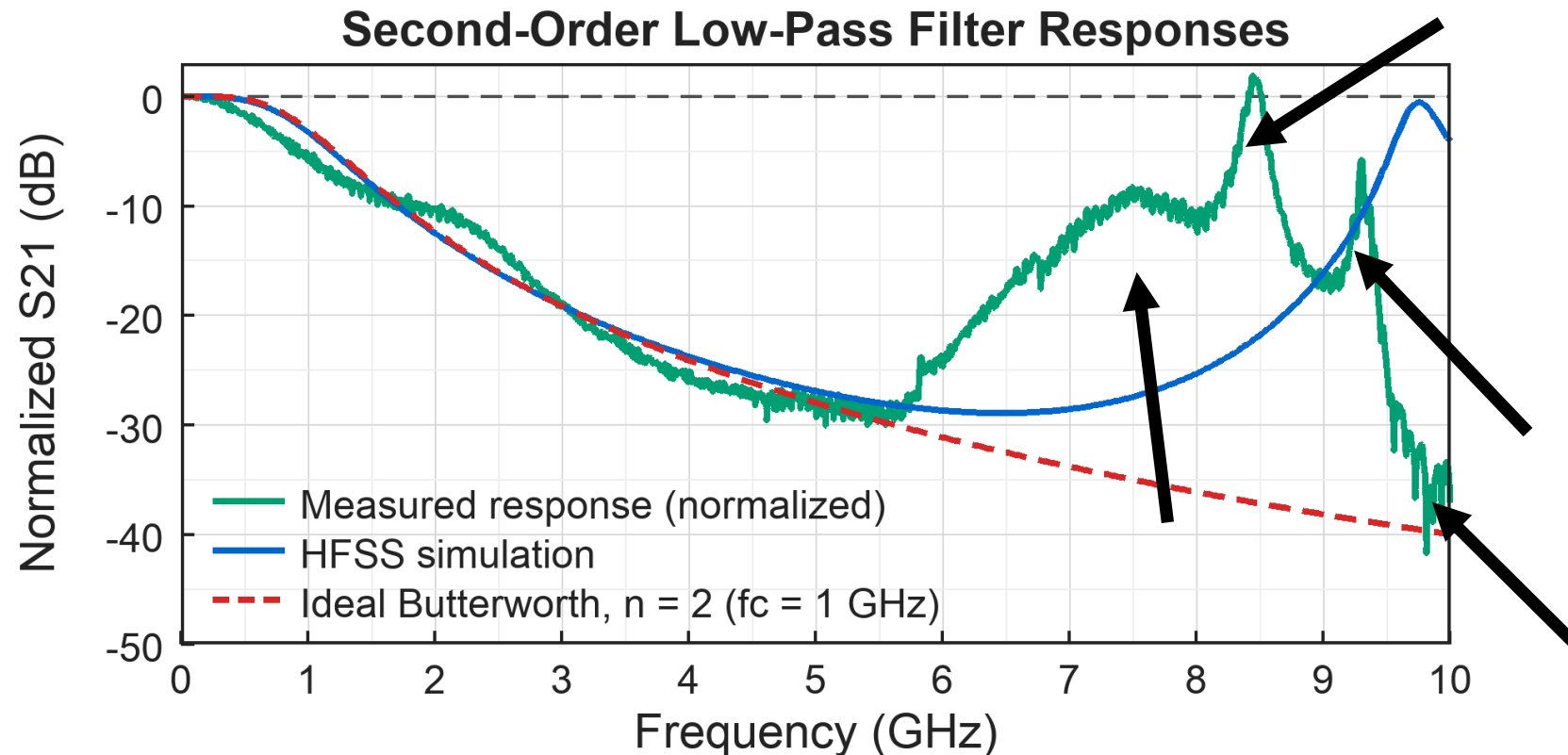
Superconductive
Environment

4 Kelvin

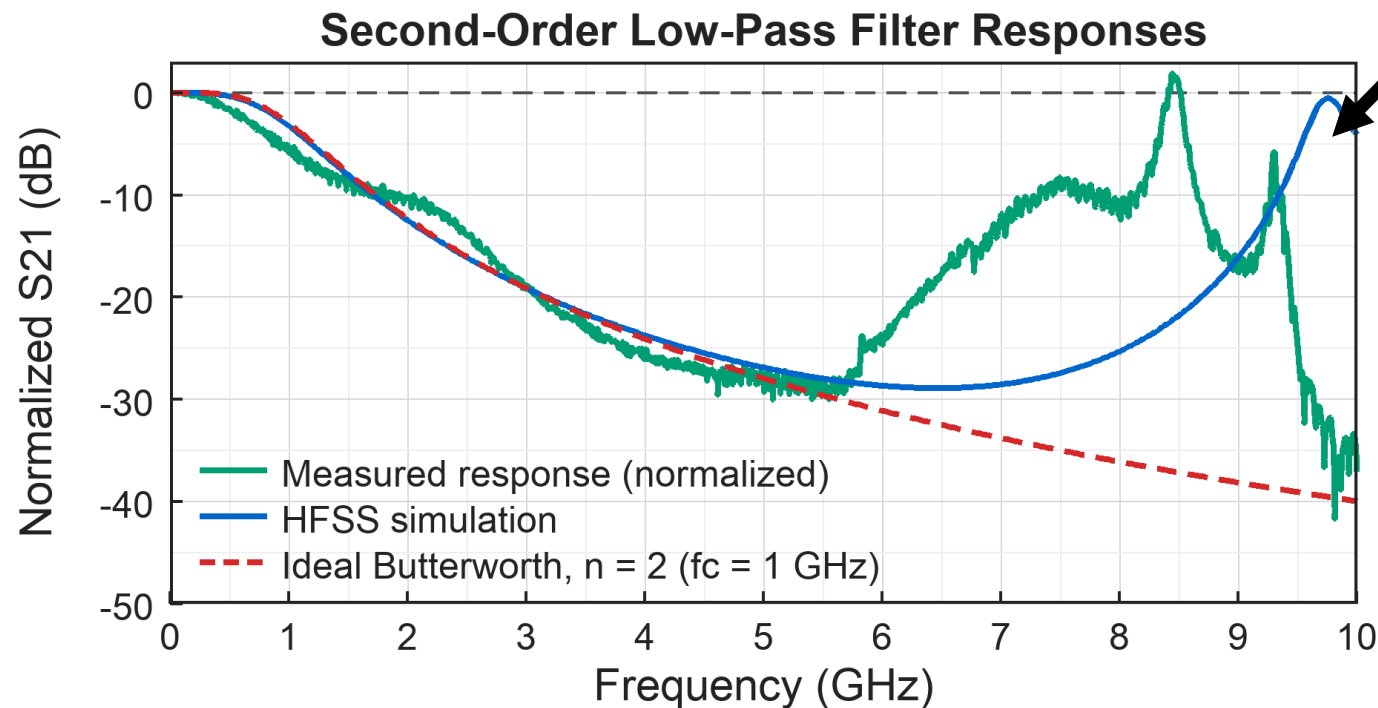


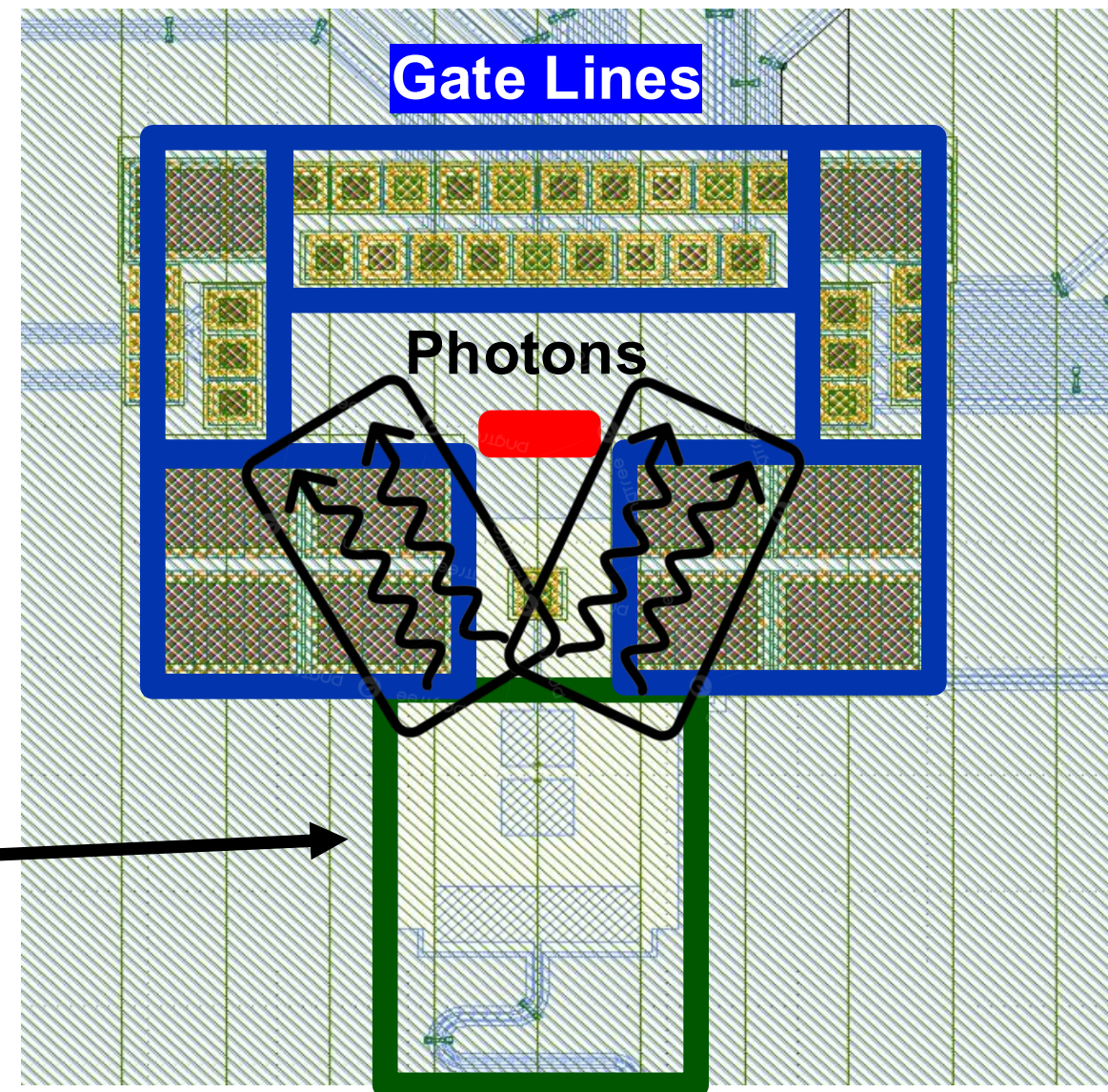
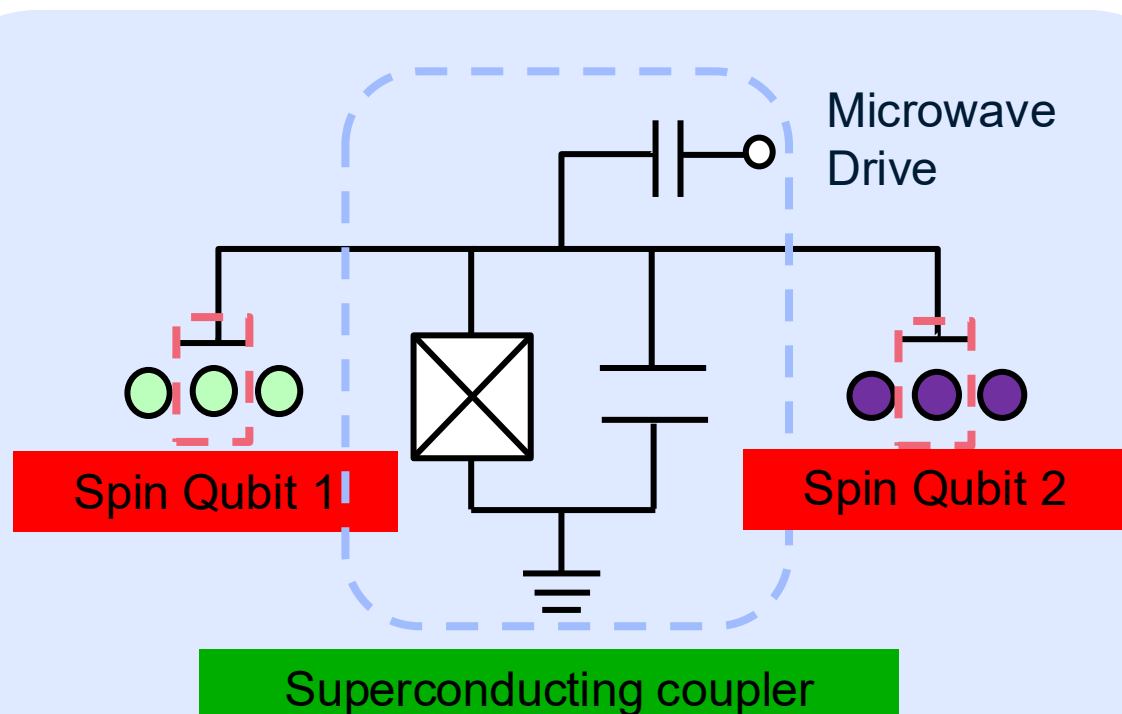


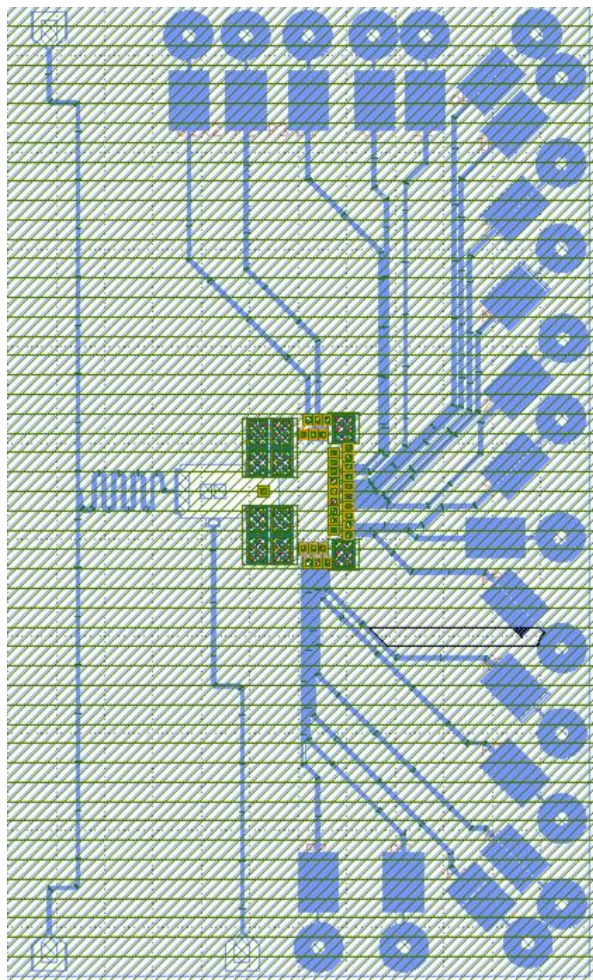
- **Potential Fabrication vs Ideal Simulation differences**
- Potential reason for the self-resonances in the stopband**



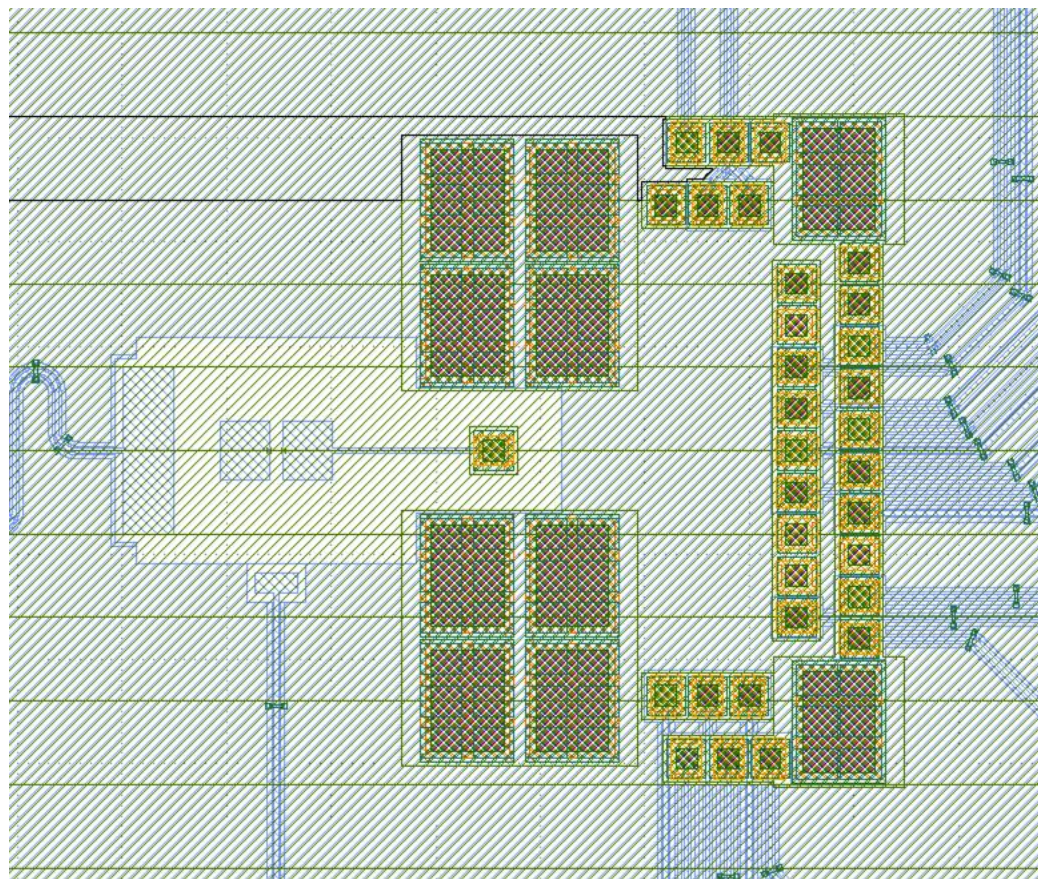
- **Impedance Boundary Discrepancies**
 - **Measurements: 50 ohm terminated on both sides**
 - **System: 50 ohm terminated on one side vs open on the other side (approaching to infinity)**
 - Leads to different structural modes



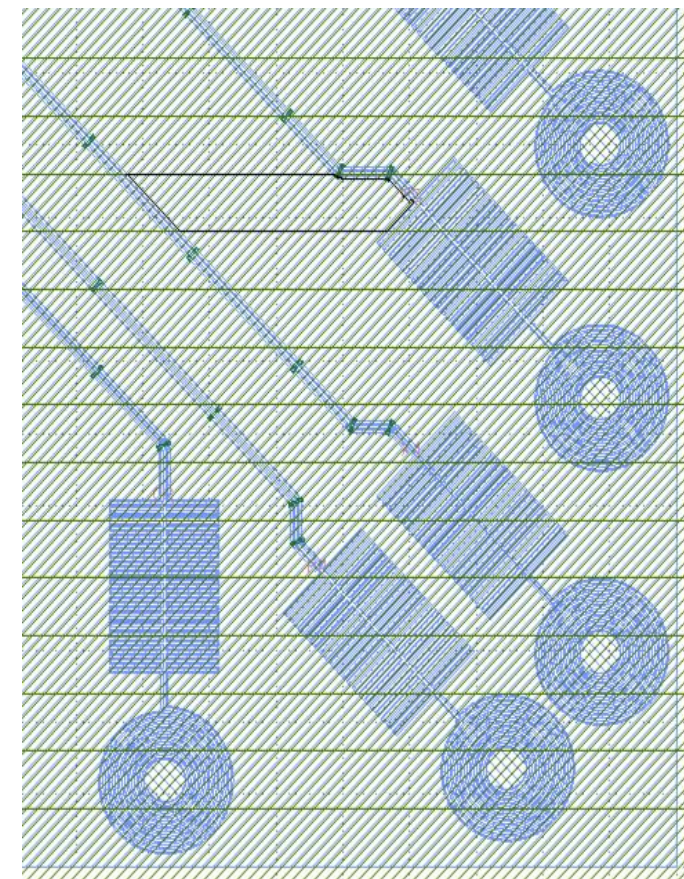




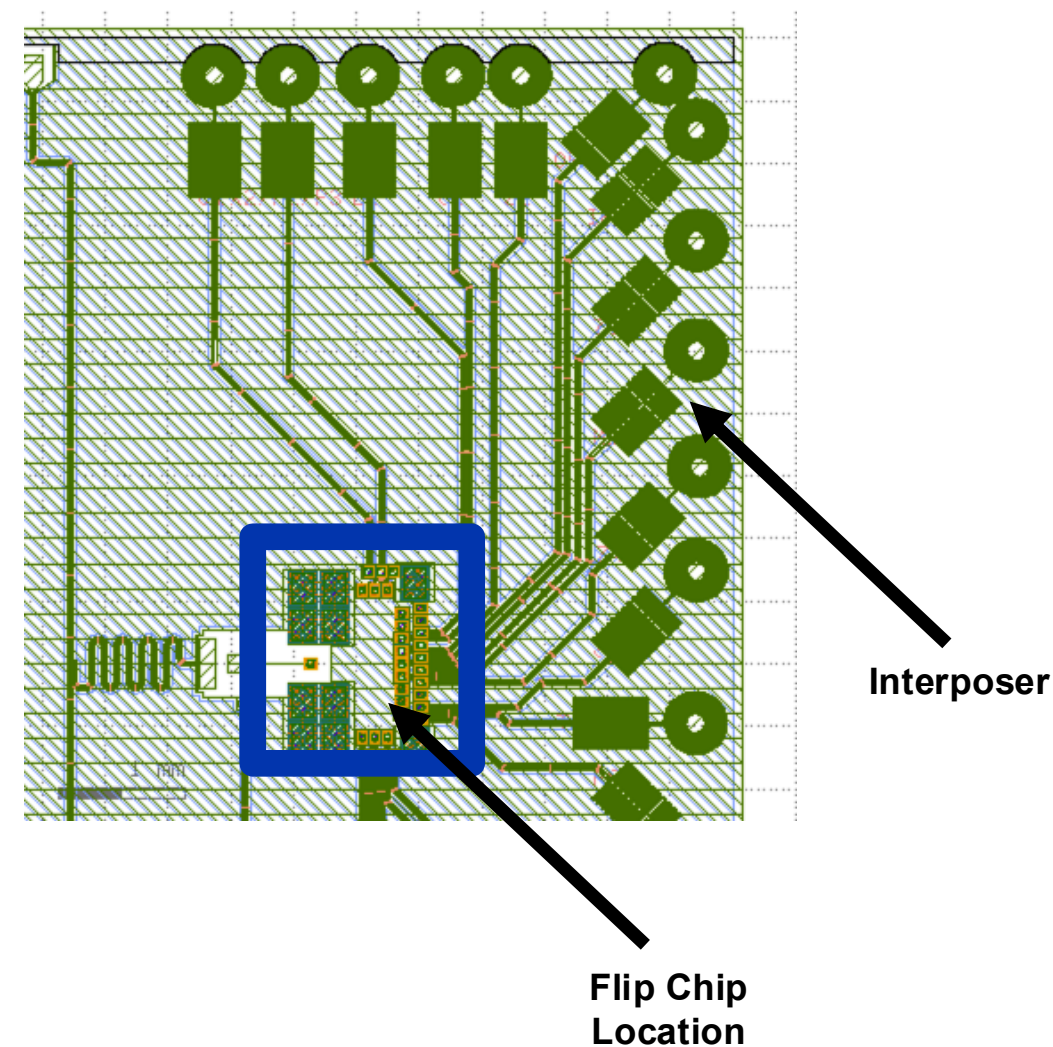
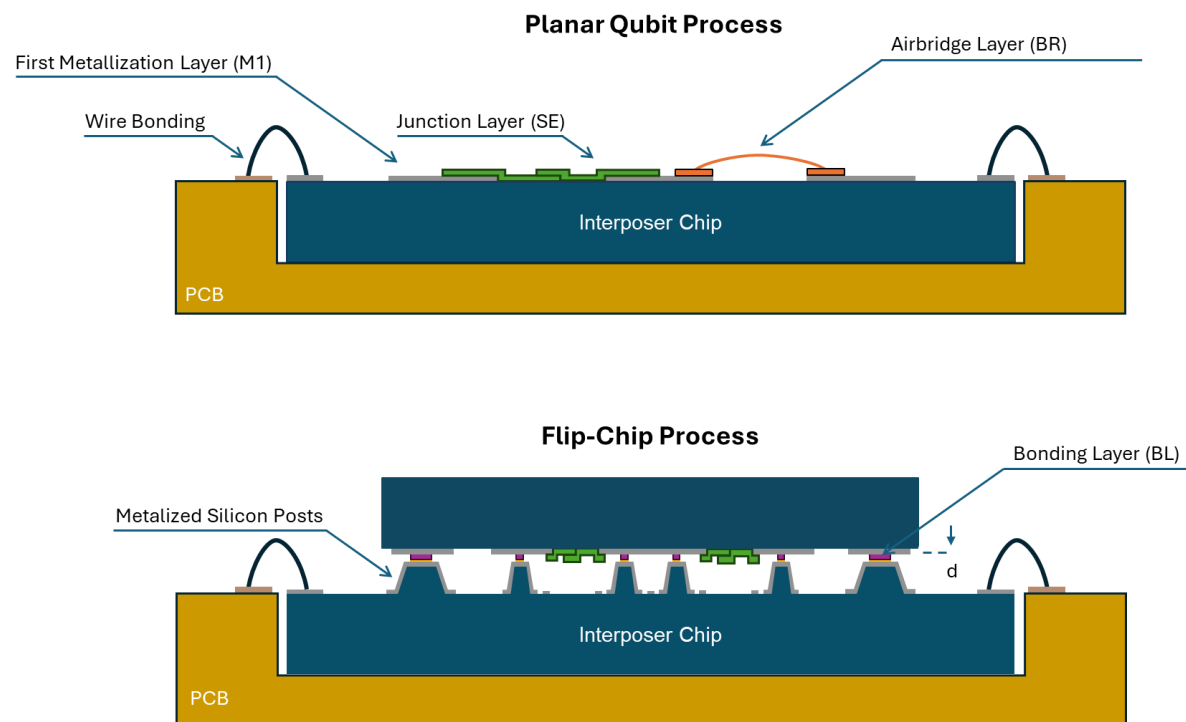
**Superconducting + Spin + Gate
lines Qubit System**

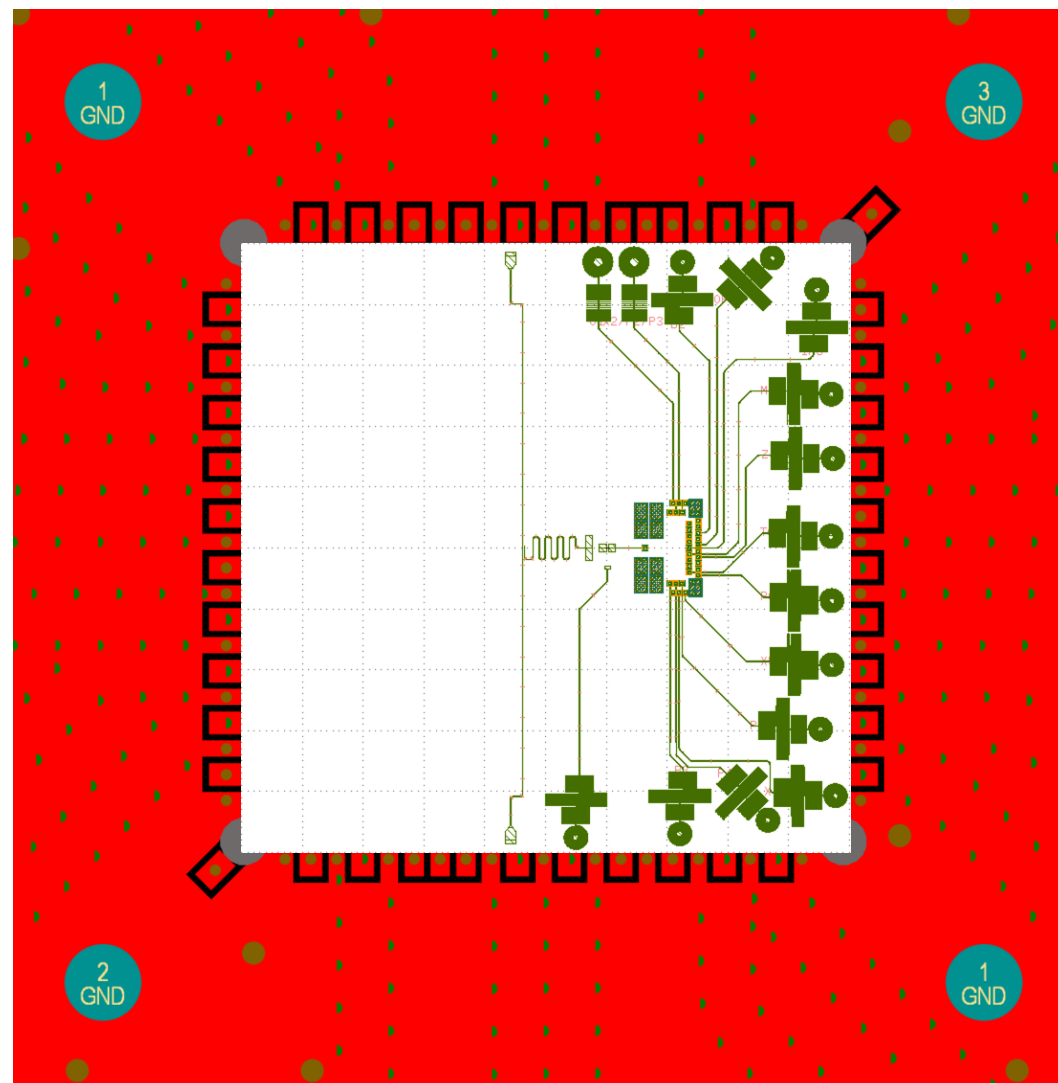


Superconducting + Spin Qubit System

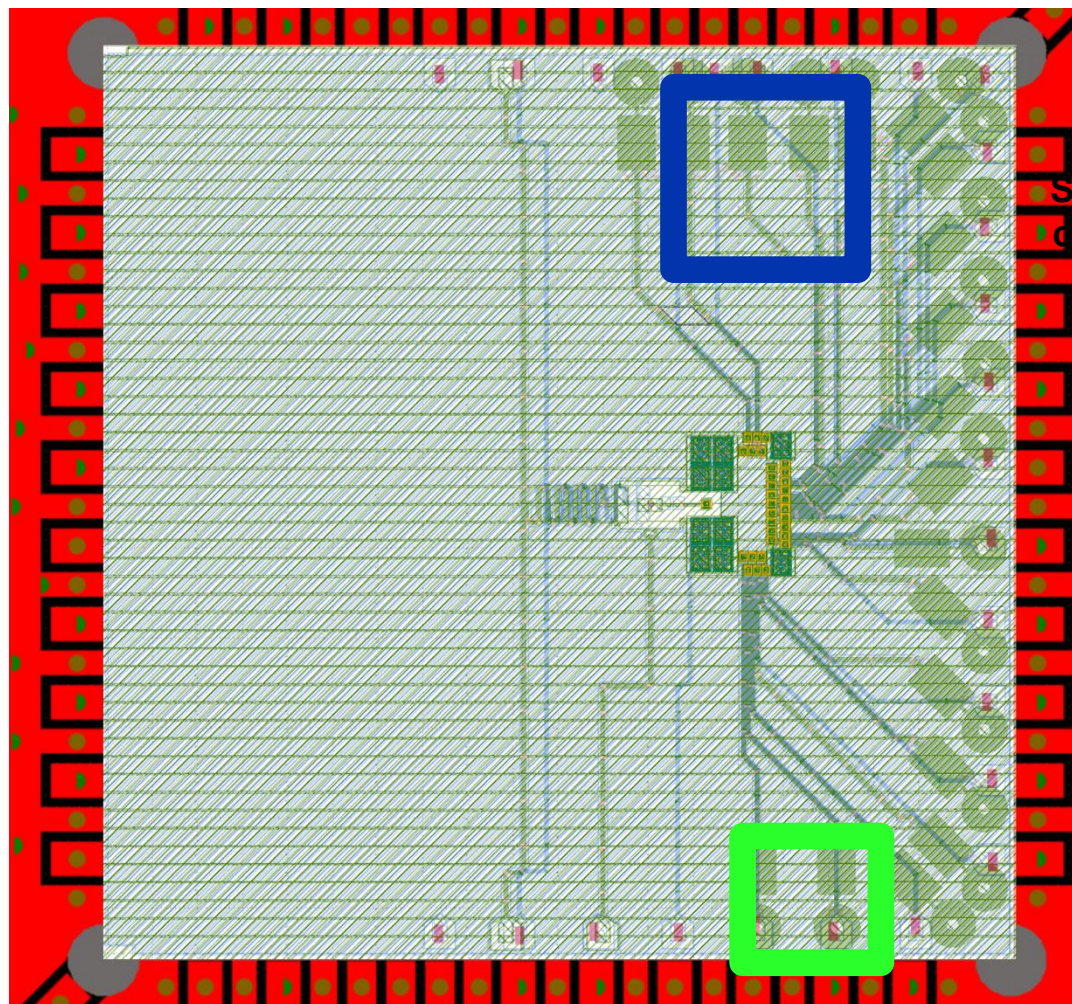


Filter Lines

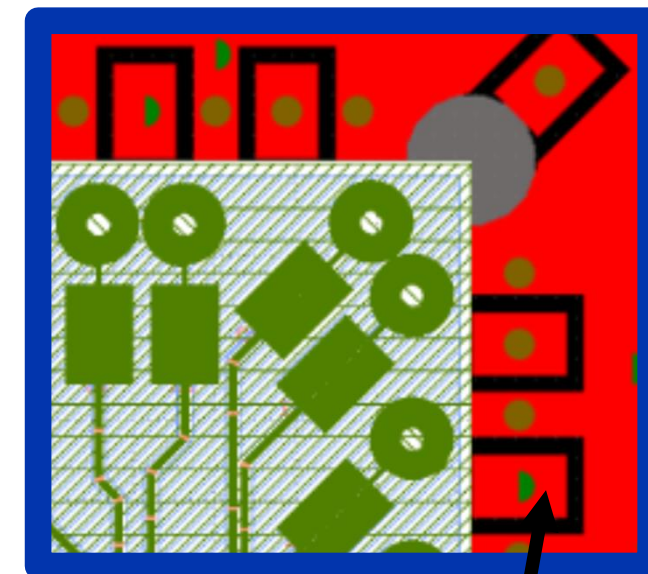




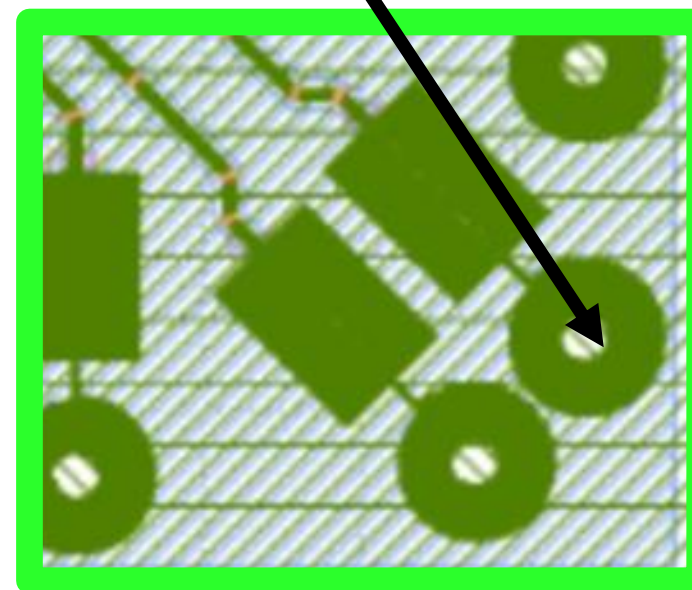
Thank you, Gabriel!



Spiral Inductor
center pad will
be used as
launchers



Bondpads



Qubit package in BF5

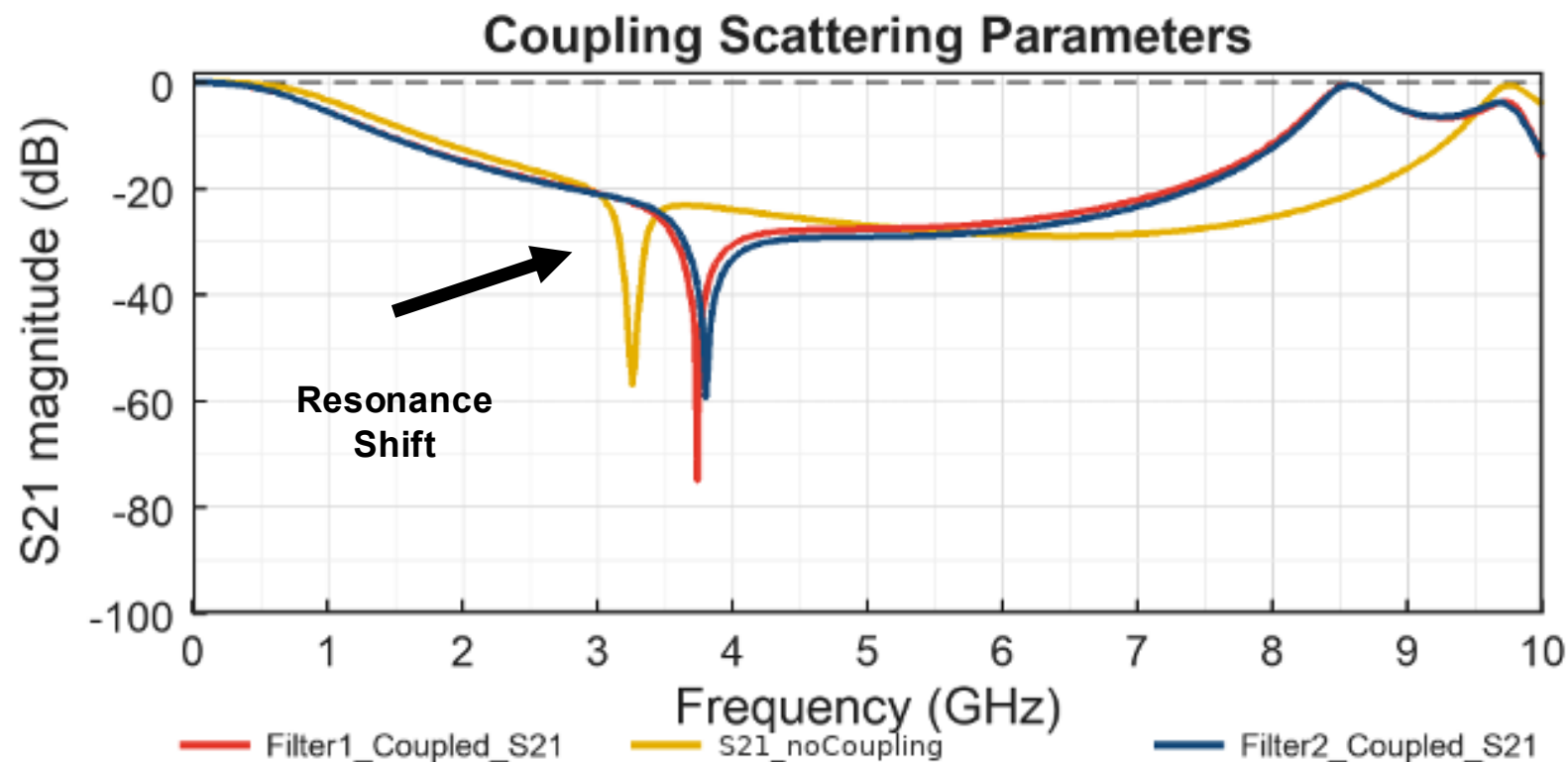
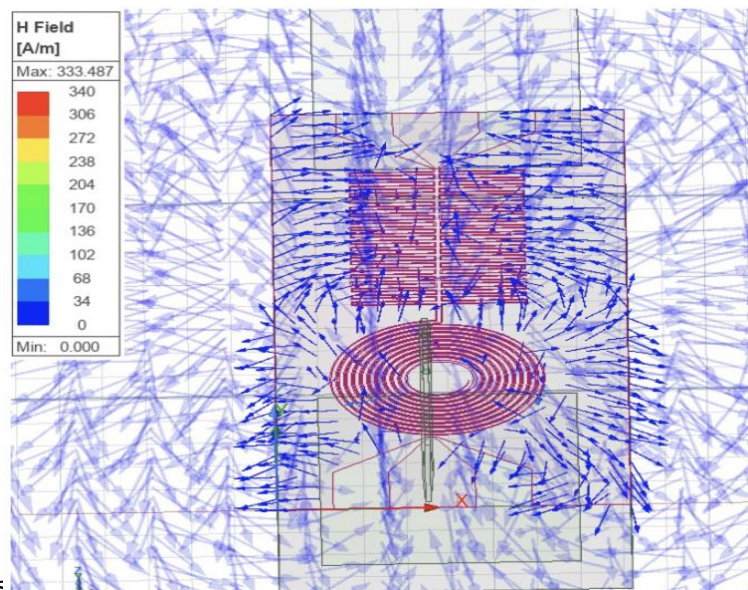
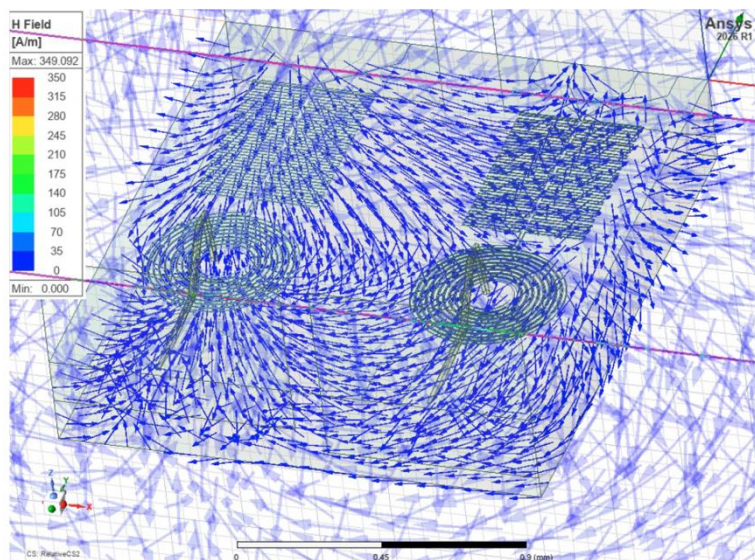
HFSS Eigenmodes from 0 to 10 GHz

Only modes satisfying $0 \leq \text{Re}(f) \leq 10$ GHz are included

Shortest Path			
Mode	Re(f), GHz	Im(f), GHz	Q
Mode 1	4.39602	3.29189e-4	6,677.05
Mode 2	8.57176	6.73890e-5	63,599.1

Longest Path			
Mode	Re(f), GHz	Im(f), GHz	Q
Mode 1	2.99880	2.60440e-4	5,757.18
Mode 2	8.02873	7.41550e-4	5,413.48
Mode 3	9.08462	1.60465e-3	2,830.72

Complex frequency: $f = \text{Re}(f) + i \text{Im}(f)$ • $Q = \text{Re}(f) / [2 \text{Im}(f)]$



Conclusions:

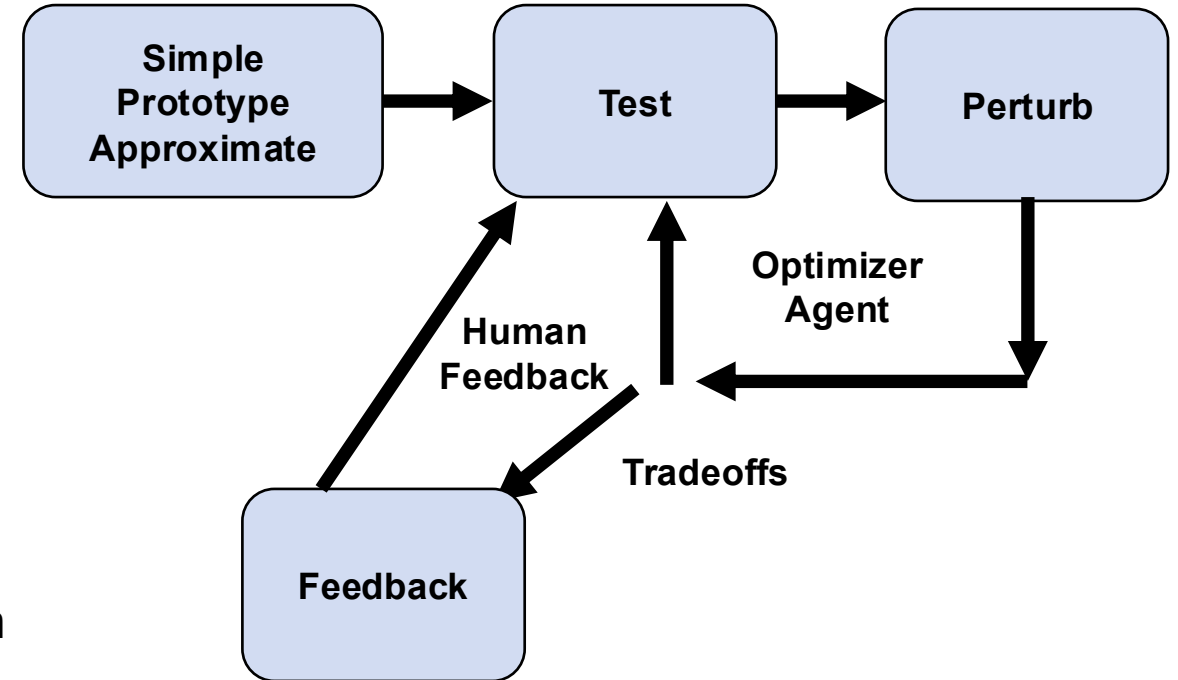
The change in resonance does seem noticeable; but, given the level of attenuation in the 5-8 GHz range is low, the effect would practically not be that significant

The process was a closed-loop HFSS optimization:
(Typical in advanced filter design)

1. **Generate a candidate geometry.**
2. **Validate fabrication and connectivity.**
3. **Generate the exact GDS.**
4. **Simulate that GDS in HFSS.**
5. **extract S11/S21.**
6. **Score cutoff, rejection, resonances, size, and fabrication compliance.**
7. **Use the measured result to select the next design**

Tradeoffs:

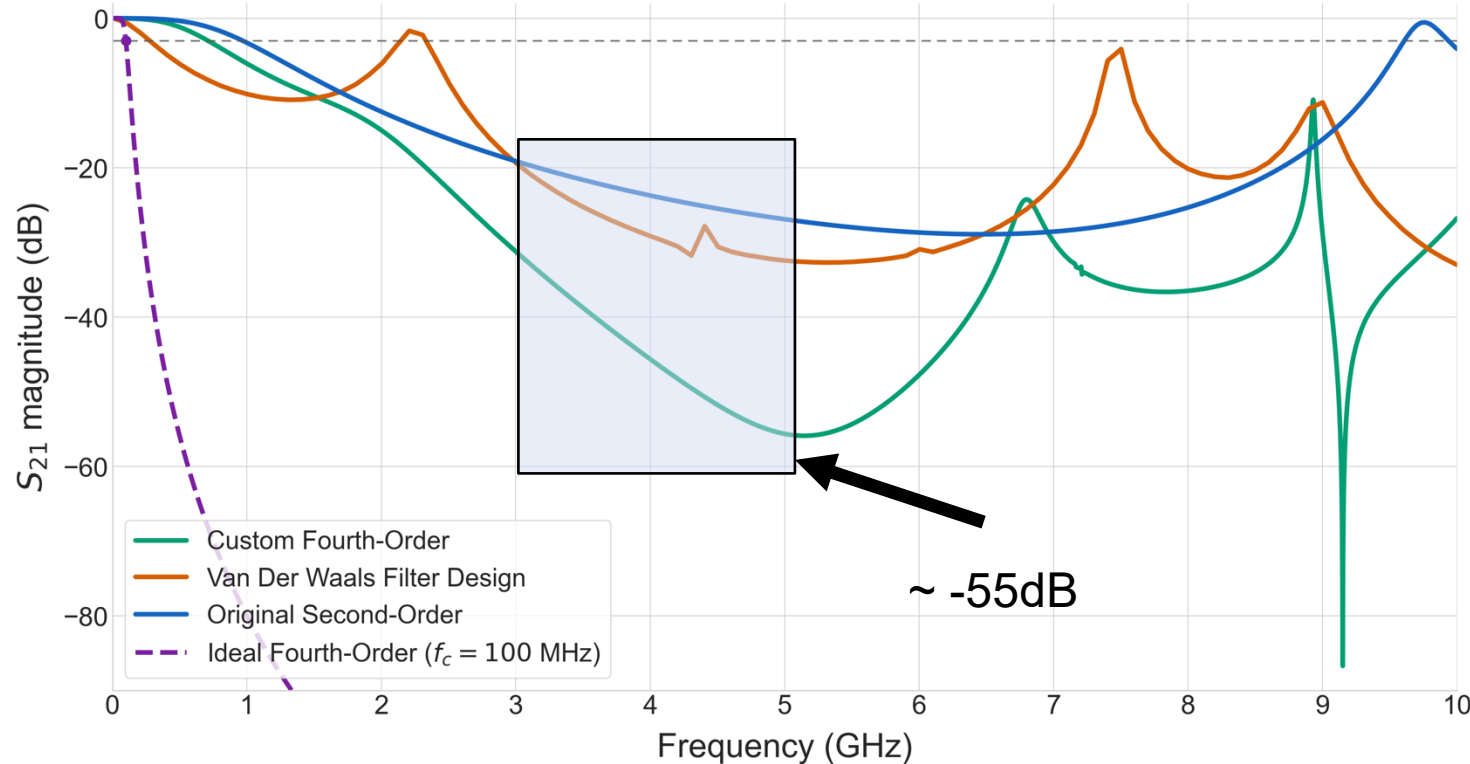
cutoff, rejection, resonances, size, and fabrication compliance
~ Transmon operation region



Follows filter design principles

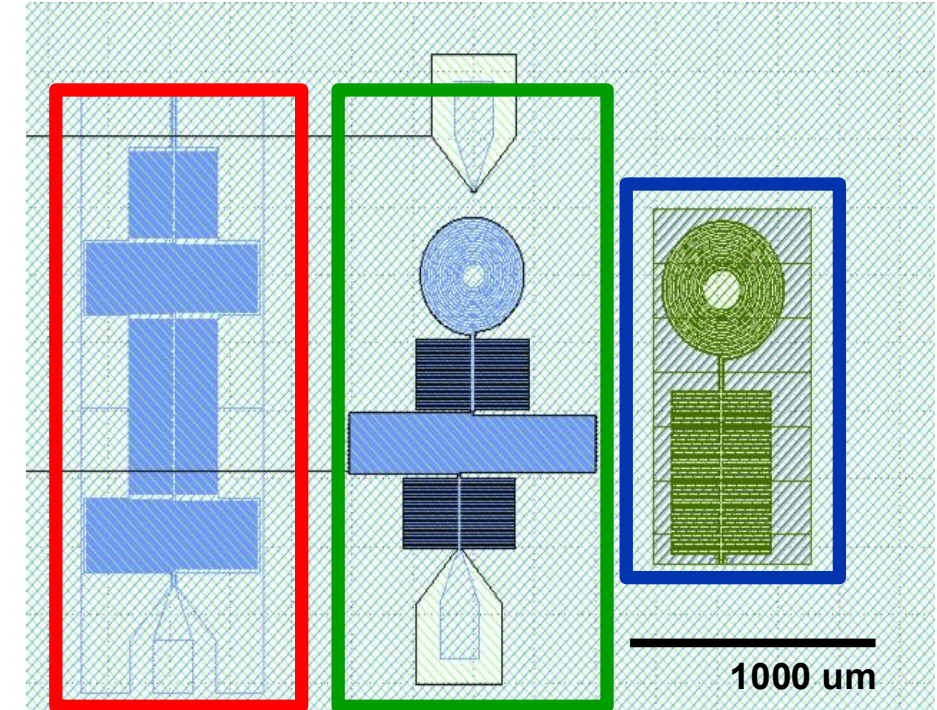
1. Model the ideal filter response
2. Select layout and synthesize circuit
3. Convert circuit values into dimensions
4. Optimize geometry using EM simulation

Low-Pass Filter Design Comparison



Far from Ideal...

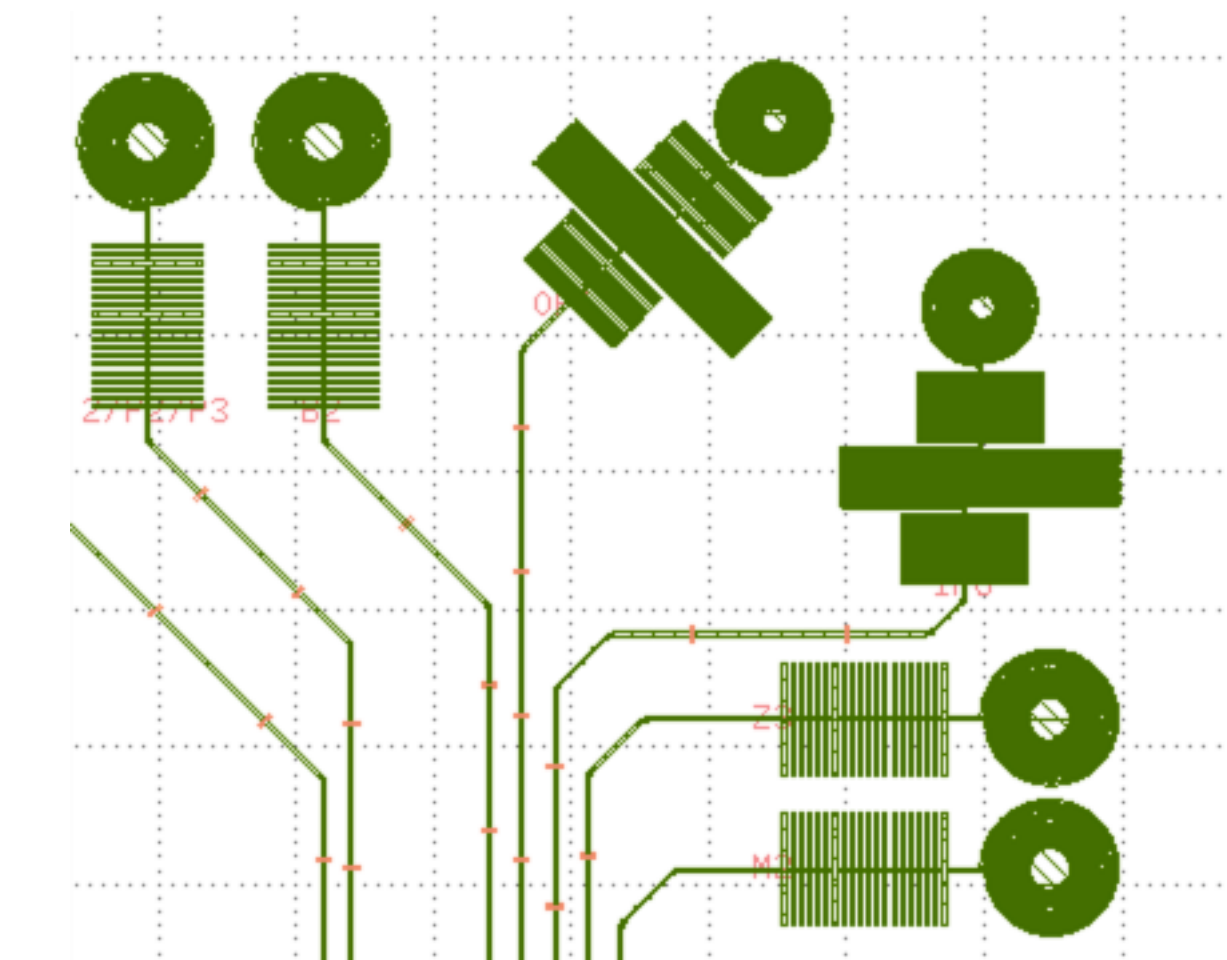
Interesting trend found:
Moving the cutoff lower increased distributed parasitic behavior. In other words, increased self-resonance



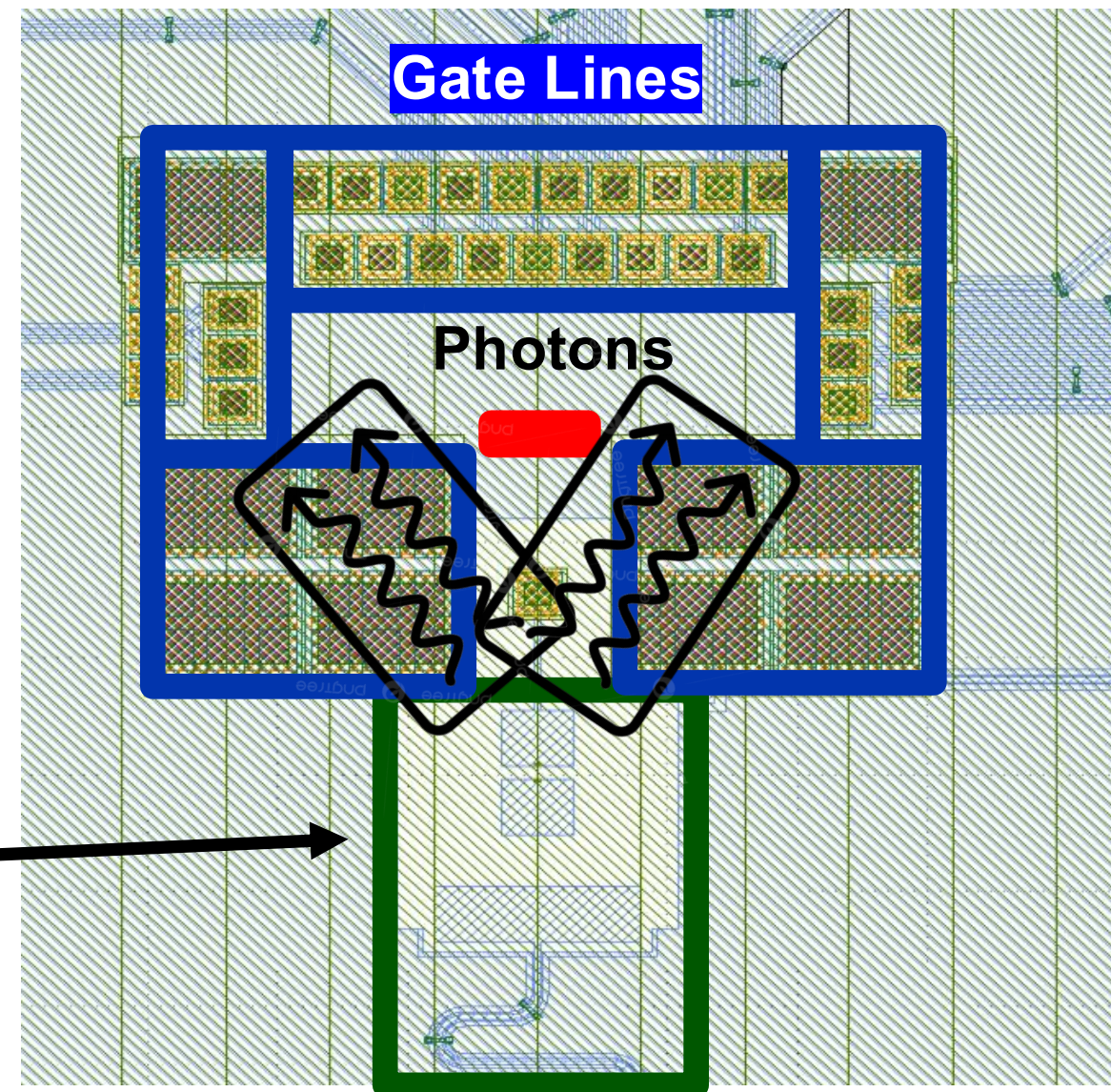
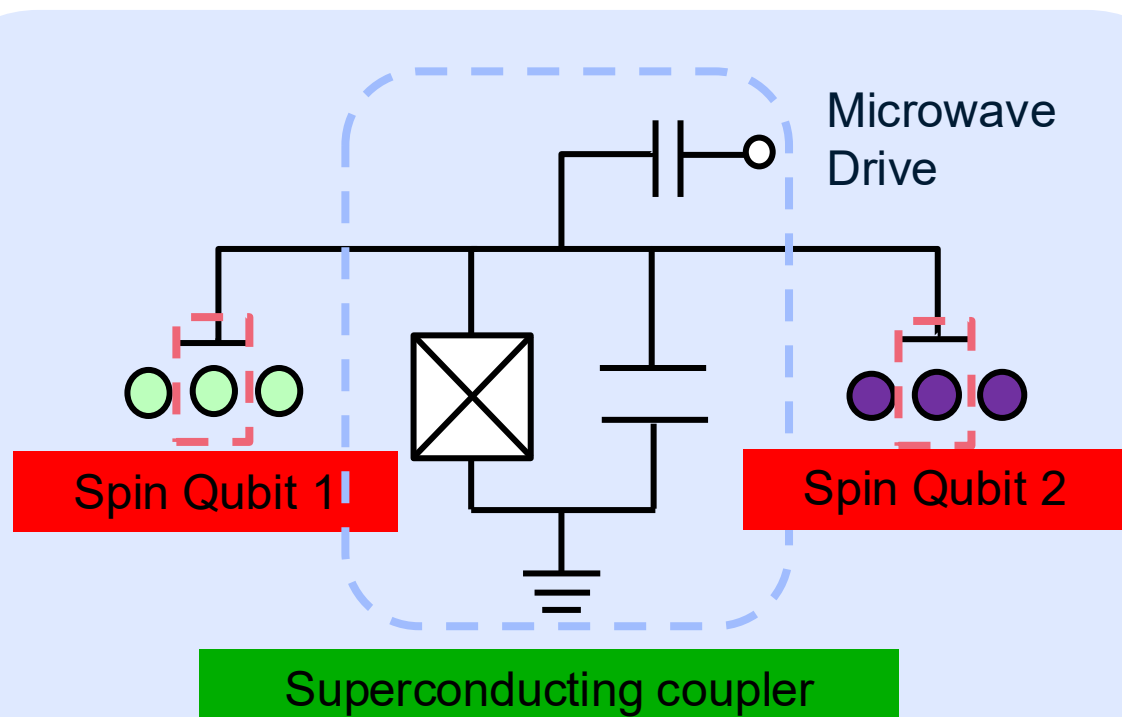
Thank you, Van Der Waals Team!

First -3 dB cutoffs:

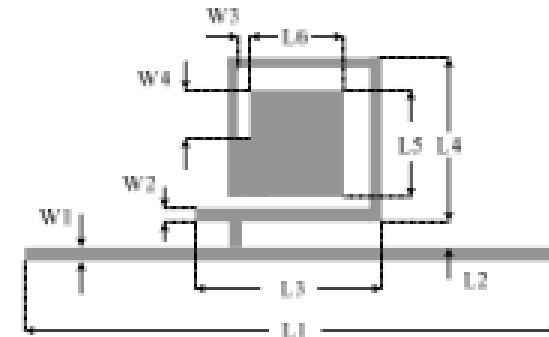
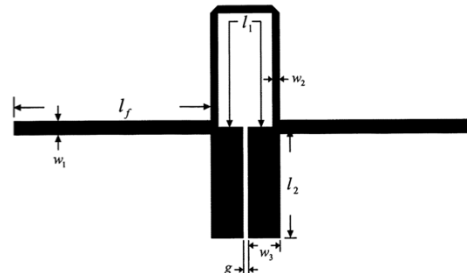
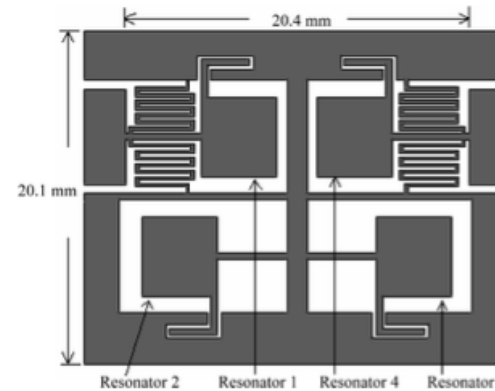
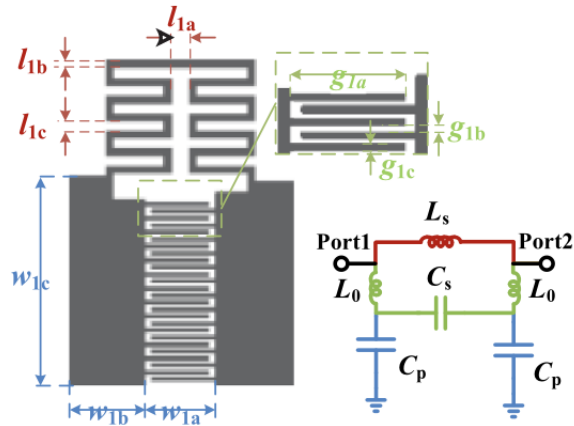
- Custom "Fourth-Order": **0.710 GHz**
- Van der waals Filter design: **0.647 GHz**
- Original "Second-Order": **0.969 GHz**



**Placed on
important gate lines
given space
constraints**



- Small dimensions (um) lead to us to focus on approx. lumped element.
Operation region leads smaller than a tenth of a wavelength
 - Reference: 10 GHz: about 11.9 mm in CPW silicon
- Higher order filters offer better chance for attenuation for a wide range (100 MHz – 10 GHz)



- **Topologies (Higher order + Compact Designs)**
- **Ensure modes don't affect filter performance**
- **Interposer Layout**
- **Fabricate new designs**
- **Optimization workflow**
 - **Program that automatically generates desired frequency Responses**

Quantum-Enabled Computation Group, MIT-LL



Superconducting Qubits Team:

Group Leaders: Mollie Schwartz, Kyle Semiak, Steven Weber, Jonilyn Yoder

Team Leads: Luke Burkhart, Rabindra Das, Michael Gingras, Thomas Hazard, Cyrus Hirjibehedin, Bethany Niedzielski Huffman, Katrina Sliwa, Donna-Ruth Yost

Team: Mike Augeri, Kate Azar, Peter Baldo, Jeffrey Birenbaum, Logan Bishop Van Horn, Tristan Brown, Gregory Calusine, Glenn Carl, Felipe Contipelli, John Cummings, David Danza, Dan Davis, Renée DePencier Piñero, Steven Disseler, Bryce Fisher, Ben Freiman, Jeffrey Gertler, Kevin Grossklaus, Karen Harmon, Mike Hellstrom, Gerry Holland, Jennifer Hritz, Sokharom Huon, Andrew J. Kerman, David Kim, Jeff Knecht, Arthur Kurlej, Duncan Miller, Nathan Miller, Jovi Miloshi, Maddie Morocco, Kevin Obenland, Christopher O'Connell, Mallika Randeria, Matt Ricci, Lila Rodgers, Robert Rood, Gabriel Samach, Meghan Schuldt, Arjan Sevi, Marcus Sherwin, Hannah Stickler, Chris Thoummaraj, Kunal Tiwari, David Volfson, Terry Weir, Zally Yun

Visiting Member: Serra Erdamar

MIT Engineering Quantum Systems (EQuS)



William D. Oliver, Jeff Grover, Max Hays, Joel Wang, Terry Orlando, Kyle Serniak, Chihiro Watanabe, Lenore Buck Simon Gustavsson (now at Google Quantum AI)

Postdocs: Aziza Almanakly, Réouven Assouly, Fabrizio Berritta, Farid Hassani Bijarbooneh, Jorge Marques, Doug Pinckney, Holly Stemp, Helin Zhang

PhD students: Vaishnavi Addala, Junyoung An, Lamia Ateshian, Kate Azar, William Banner, Cora Barrett, Frederike Brockmeyer, Shoumik Chowdhury, Gabriel Cutter, Andy Ding, Ben Helman, Kotaro Hida, Shantanu Jha, Om Joshi, Harry Kang, Gyunghun Kim, Junghyun Kim, Jonathan Knoll, Chris McNally, Maggie Marte, Miguel Moreira, Sarah Muschinske, David Pahl, Lukas Pahl, David Rower, Chia-Chin Tsai, Hung-Yu Tsao, Beatriz Yankelevich, Sameia Zaman

Undergraduates: Oswin Cervantes, Justin Chen, Matan Yablon

Visiting Member: Daniel Rocha, Gabriele Roller

Collaborators: Aksh Dogra, Joe Formaggio, Kevin O'Brien (MIT); Ben Loer (PNNL)

HFSS Eigenmodes from 0-10 GHz

Only modes satisfying $0 \leq \text{Re}(f) \leq 10$ GHz are included

Shortest Path · 1378.334 μm			
Mode	Re(f), GHz	Im(f), GHz	Q
Mode 1	0.13951	1.83887e2	0.500
Mode 2	0.67905	1.70914e-1	2.048
Mode 3	8.29290	1.19127e-1	34.811

Longest Path · 4622.897 μm			
Mode	Re(f), GHz	Im(f), GHz	Q
Mode 1	0.60098	1.71067e-1	1.826
Mode 2	6.34814	5.65714e-3	561.073
Mode 3	8.43758	1.37402e-1	30.708

Complex frequency: $f = \text{Re}(f) + i \text{Im}(f)$ • $Q = |f|/[2|\text{Im}(f)|]$